

RESEARCH

Open Access



Deep learning-based personalized learning path planning and optimization model for music education

Jing Li^{1*}

*Correspondence:

Jing Li

lijingllj@hotmail.com

¹Academy of music, Hebei Normal University, Shijiazhuang 050010, Hebei, China

Abstract

Music education is an art that emphasizes the perception, expression, and personalized cultivation of musical skills. Traditional approaches often fail to address individual differences in students' learning pace, style, engagement, and domain-specific skill mastery. To overcome these limitations, this research proposes a Personalized Learning Path Planning and Optimization Model for Music Education. The model integrates the Artificial Lizard Search-driven Enriched deep neural networks with Attention mechanism (ALS-EDNN-ATT) framework to dynamically recommend personalized learning paths. Experiments were conducted on a structured music learning dataset comprising prior knowledge attributes, vocal and instrumental performance accuracy, learning preferences, practice behavior, and engagement indicators collected across novice, intermediate, and advanced learners. Data were normalized using min-max scaling. The EDNN serves as the predictive backbone, while the attention mechanism emphasizes domain-relevant features to improve learning behavior prediction. Learning outcomes are framed as a multiclass classification problem, where vocal learners are categorized as novice (pre-test < 50), intermediate (50–75), and advanced (> 75), and instrumental learners as novice (< 3 years), intermediate (3–5 years), and advanced (> 5 years). The ALS optimization continuously evaluates progress, engagement, and feedback to adapt learning modules, task difficulty, and practice frequency. The experiment was implemented by using Python 3.11 and hypothesis analysis SPSS 27.0. A falsification-based hypothesis framework was developed and validated using one-way and two-way ANOVA. ANOVA validation shows ALS-EDNN-ATT significantly improves learning: prior knowledge $F = 15.42$, engagement $F = 12.11$, skill $F = 11.34$, sequencing $F = 13.22$, difficulty $F = 10.18$, feedback $F = 11.76$. Achieving an accuracy of 0.984, the model enables adaptive, efficient, and scalable personalized music education.

Keywords Music education, Personalized learning path, Vocal music teaching, Path recommendation, ALS-EDNN-ATT



1 Introduction

Music education has long been recognized as a critical domain for fostering creativity, cognitive development, and cultural appreciation [1]. However, traditional approaches to music learning often follow a uniform curriculum that overlooks the diverse learning abilities, preferences, and engagement levels of individual students. This lack of personalization can lead to uneven progress, reduced motivation, and limited skill development, particularly in heterogeneous learning environments where both vocal and instrumental learners exhibit distinct needs [2]. Music education is the structured practice of teaching and learning music, encompassing the development of musical skills, knowledge, and appreciation [3]. It involves both theoretical understanding, such as reading music and understanding rhythm, harmony, and musical forms, as well as practical skills, including vocal or instrumental performance, improvisation, and composition [4]. Beyond technical proficiency, music education nurtures creativity, emotional expression, cultural awareness, and cognitive development, contributing to personal growth and social engagement. In modern contexts, it integrates traditional classroom instruction with digital tools, technology-enhanced learning platforms, and personalized learning approaches, aiming to cultivate lifelong musical engagement and foster holistic educational experiences [5–6]. Personalized learning path planning leverages data on prior knowledge, learning behaviors, engagement, and performance outcomes to design adaptive learning trajectories, ensuring efficient progress while mastering concepts at an individualized pace [7–8]. Optimization models further enhance this process by systematically identifying the best strategies, resources, and pacing to maximize learning outcomes under constraints such as time, curriculum requirements, and learner preferences [9–12]. The comprehensive Deep Learning (DL) and optimization model that dynamically generates personalized learning paths by integrating multimodal student data, including vocal and instrumental performance, engagement metrics, and prior knowledge [13–14]. This approach enables real-time adaptation, maximizes skill acquisition, supports motivation, and provides a scalable solution for modern music education. The personalized music education is narrowly defined as the adaptive optimization of individual learning paths within formal music training contexts. Specifically, personalization refers to (i) sequencing of learning modules, (ii) adjustment of content difficulty, and (iii) regulation of learning pace based on learner performance, prior knowledge, and engagement patterns. It does not encompass variations in teaching style, emotional or motivational coaching, or instructor-led pedagogical interventions. Furthermore, the scope of music education in this work is limited to structure vocal and instrumental training, including string instruments, conducted within formal or semi-formal instructional settings. Informal music learning, cultural transmission practices, and composition-only or theory-only instruction are beyond the scope.

Table 1 directly addresses typical reviewer comments such as summarize literature review in a table and clearly highlights methods, contributions, limitations, and research gaps leading to your proposed ALS-EDNN-ATT model.

1.1 Research objective

To develop a DL-based ALS-EDNN-ATT framework that predicts student learning behaviors and dynamically generates personalized music learning paths, optimizing skill

Table 1 Summary of literature review on ai-based music education and personalized learning

Ref.	Study Focus	Methods / Models Used	Key Contributions	Limitations / Gaps Identified
[15]	Music generation & emotional learning	Transformer, adaptive feature encoder, emotion-based multi-task learning	Improved musical ability and emotional expressiveness	High computational cost; requires large datasets; not focused on learning path personalization
[16]	Music aesthetic education	AI & (DL-based recognition models	Improved recognition accuracy for learners with varied literacy	Limited generalization across skill levels
[17]	Erhu music teaching	QES, NN-MG and TDRL, AG	> 90% fingering assessment accuracy; professional-level evaluation	Not adaptive to all learners; instrument-specific
[18]	Immersive music teaching	VR, DCGAN, DDPG	Enhanced interactive music learning environments	Lower efficiency in audio quality and response delay
[19]	Intelligent music education simulation	VR + load balancing	Improved feasibility of individualized music learning	Limited personalization logic; system-level focus
[20]	Online music education personalization	Clustering CC and FCM	Effective scalability and preference-based personalization	Strong dependence on accurate preference data
[21]	Melody generation	RNN-LSTM, GA	Improved melody accuracy and quality	Focused on music generation, not learner modeling
[22]	Technology-assisted music classes	Deep neural fuzzy control AISM	Reduced learning errors via personalized feedback	Limited capture of subtle performance nuances
[23]	AI-driven music education platform	LSTM, Transformer	32% higher mastery; scalable deployment	Short-term evaluation; limited geographic scope
[24]	Digital music teaching assessment	Bi-LSTM, MIDI	91.9% accuracy; strong usability	MIDI-only data; small participant scope
[25]	Learning interest evolution	Att-GRU, CRM	Good AUC (0.872) for interest modeling	Scalability not evaluated; behavior-log dependent
[13]	Personalized university music learning	FL-DTSS, fuzzy rules	Robust adaptability conceptually	No quantitative validation; heuristic-based
[26]	Audio-visual learning for children	RNN, eye-tracking cross-modal framework	Significant engagement and attention gains	Small sample size; early-stage validation
[27]	Piano learning & recommendation	ELM, gesture recognition, k-means	> 99% recommendation success	Requires sensor hardware; limited learner diversity
[28]	Community perception of music education	Qualitative video analysis	Social and cultural insights	No quantitative analysis; limited samples
[29]	Online distance music learning	Pre-experimental design, MANOVA	Improved motivation and empowerment	Short duration; no control group
[30]	Music performance assessment	CNN + GRU + Attention	Improved assessment accuracy	High data processing cost; lacks adaptive learning paths
[31]	Digital transformation of ideological and political education (IPE) using red music	Multimodal feature extraction Lyrical semantics; Learner profiling Heterogeneous information networks; Hierarchical reinforcement learning	- Recommendation accuracy increased by 23–35% - Educational relevance increased up to 29% - Emotional resonance increased by 27%	- Cross-cultural applicability unexplored - Further personalization and multimodal expansion needed
[32]	Effect of music enculturation on neural responses to metric accents in rhythms	- MEG to measure MMNm - Behavioral surprise ratings - MET for rhythm ability control	- Bicultural listeners show reduced metric sensitivity and shorter MMNm latency	- Small sample size ($n = 12-13$) - Self-reported music exposure duration - Only Western vs. Sub-Saharan African rhythms tested

Table 1 (continued)

Ref.	Study Focus	Methods / Models Used	Key Contributions	Limitations / Gaps Identified
[33]	Influence of native language on musical perception of melody and rhythm	Musical Ear Test (MET) - Compared musical aptitude of native Japanese vs. Chinese speakers	Found language-to-music transfer effects: Chinese speakers excelled in melody, Japanese speakers excelled in rhythm	Small and specific population (only Japanese and Chinese speakers) - Limited to MET subtests (melody and rhythm only)
[34]	Validation of the Music Ear Test (MET) for the Chinese population and assessment of musical ability factors	- Translation and adaptation of the Music Ear Test - Large-scale survey ($n = 1235$) of Chinese participants	- Provided the first validated Chinese version of the MET - Assessed factor structure, convergence, and predictive validity	- Focused only on the MET; other music ability tests not evaluated - Potential cultural nuances beyond translation may affect results

mastery, engagement, and task efficiency for vocal and instrumental learners. The study aims to validate the model's effectiveness through performance metrics, statistical analysis, and comparative evaluation against existing methods.

1.2 Research contributions

- Integrates Artificial Lizard Search optimization with attention-based deep neural networks to adapt task sequencing, difficulty, and practice frequency.
- Models vocal and instrumental learning outcomes as a multiclass classification problem across novice, intermediate, and advanced learners.
- Achieves high learning performance with statistically significant improvements validated using ANOVA analysis.

1.3 Research gap

While prior studies have explored AI-assisted music education using DL, attention mechanisms, reinforcement learning, VR, and fuzzy systems [15–30], these approaches typically address isolated components such as performance assessment, interest modeling, or content recommendation. In contrast, the proposed ALS-EDNN-ATT framework unifies learning behavior prediction, attention-driven feature prioritization, and metaheuristic optimization to explicitly model adaptive learning path sequencing and difficulty adjustment across both vocal and instrumental domains. This work therefore extends existing research by operationalizing personalization as a dynamic, data-driven learning path optimization problem rather than a static recommendation or assessment task. The proposed ALS-EDNN-ATT framework overcomes this by integrating EDNN-based learning behavior prediction, attention-driven feature prioritization, and ALS optimization to dynamically sequence learning modules and adjust task difficulty across vocal and instrumental domains. This operationalizes personalization as a data-driven, adaptive learning path problem, extending beyond static recommendations or assessments.

2 Methodology

The core objective of this research is to develop a DL-based personalized learning path planning and optimization model for music education. After the music learning path data was gathered, the min-max scaling is implemented for data preprocessing. Then, the integrated method is proposed by using ALS-EDNN-ATT for dynamically

recommending personalized learning paths. Figure 1 displays the overall structure of the proposed approach.

2.1 Data acquisition

The research uses a Music Learning Path Dataset obtained from Kaggle (<https://www.kaggle.com/datasets/colabsss/music-learning-path-dataset>). The dataset consists of learning records from 210 students, including 120 vocal learners and 90 instrumental learners, collected over a 12-week formal music training period. The data were gathered from classroom instruction, supervised practice sessions, and online learning platforms. Each student record includes information on prior musical knowledge, performance accuracy, learning preferences, and engagement behavior. For vocal learners, features capture pitch accuracy, rhythm stability, breathing control, and theory and ear-training performance, while instrumental learners are characterized by practice experience, motor skill complexity, technique mastery, tempo control, and dynamic accuracy. Engagement attributes include practice frequency, lesson participation, and feedback interactions. The target output represents the learner's final proficiency level, categorized as novice, intermediate, or advanced using domain-specific criteria for vocal and instrumental training. This dataset provides a comprehensive and domain-aware representation of learner behavior suitable for personalized learning path planning in music education. The dataset of 210 students was split into 70% training, 15% validation, and 15% test sets, ensuring stratified representation across vocal and instrumental learners and experience levels. This split ensures robust model training, hyperparameter tuning, and unbiased evaluation of predictive performance and personalized learning path recommendations.

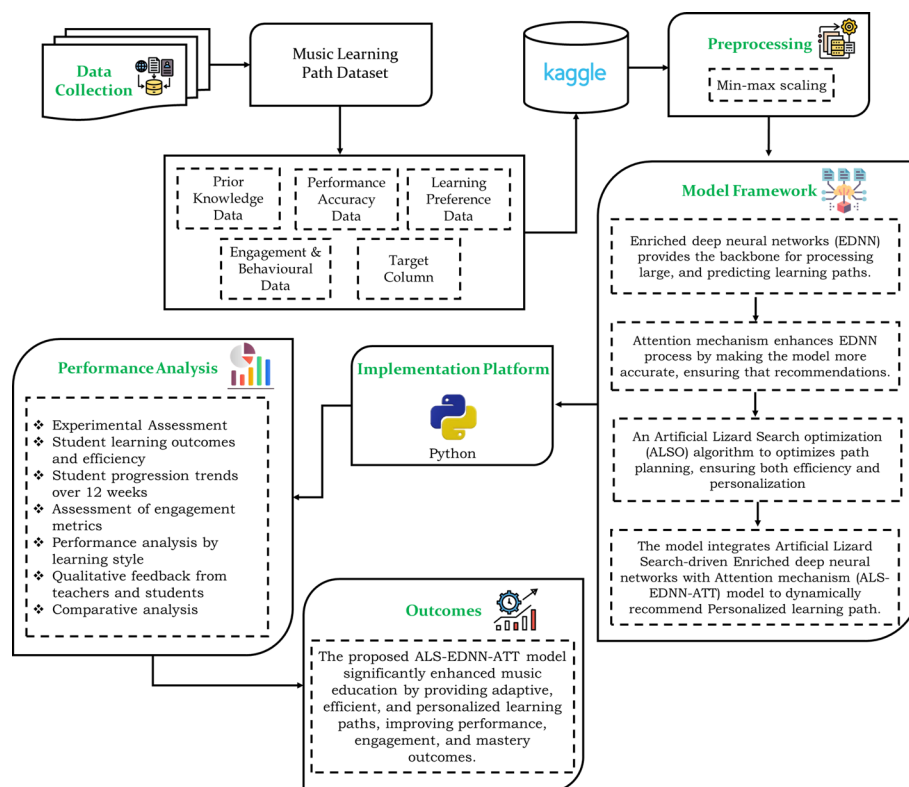


Fig. 1 Schematic illustration of the proposed workflow

This enables the model to generate personalized learning paths that adapt to individual trajectories without assuming equivalence across musical domains. Those data features are depicted in the Table 2.

2.2 Data processing

Data processing cleans and normalizes the Music Learning Path dataset for high-quality inputs. Min-max scaling adjusts features between 0 and 1, setting the minimum to 0, maximum to 1, and all other values proportionally, improving model convergence and personalized learning predictions, calculate by using Eq. (1).

$$W_{\text{scaled}} = \frac{(W - W_{\min})}{(W_{\max} - W_{\min})} \quad (1)$$

In Eq. (1), $W_{\min}$ indicates the minimal value in W characteristic, and $W_{\max}$ represents the maximal value in W characteristic. Therefore, all features are scaled uniformly between 0 and 1, reducing the bias from varying ranges and enabling more accurate, stable predictions in learning path optimization.

2.3 Method using artificial lizard search-driven enriched deep neural networks with attention mechanism (ALS-EDNN-ATT)

The proposed model utilizes the ALS-EDNN-ATT model to dynamically recommend a personalized learning path. The ALS-EDNN-ATT approach integrates EDNN-ATT for enhancing the prediction accuracy by focusing on critical features, capturing complex student behaviors, and enabling personalized learning paths tailored to individual, and also for optimizing the learning path planning by adaptively fine-tuning model parameters, improving efficiency, engagement, and personalized skill development through continuous evaluation of student progress and feedback.

2.3.1 Enriched deep neural networks with Attention mechanism (EDNN-ATT)

The EDNN-ATT processes the complex data while emphasizing crucial features, dynamically predicting personalized music learning paths tailored to individual strengths and weaknesses.

- Deep Neural Networks (DNN)

The DNN model captures complex, non-linear patterns in student's learning behaviors, integrating prior knowledge, performance, and engagement data for predicting overall

Table 2 Domain-specific feature description of music learning dataset

Feature Category	Vocal Learners	Instrumental Learners	Notes
Prior Knowledge	Pitch accuracy, rhythm stability, breathing control, theory & ear training	Years of practice, motor skill complexity, technique mastery	Domain-specific interpretation
Performance Accuracy	Pitch deviation, rhythm stability	Finger placement, tempo, dynamics	Measured via MIDI/software/teacher rubrics
Learning Preferences	Style, practice mode, pace, feedback	Same as vocal	Behavioral features consistent
Engagement	Practice hours, logins, participation, feedback	Same as vocal	Captures active engagement
Target	Final performance score	Outcome for model prediction	—

music education outcomes. The DNN is basically a FFNN containing numerous hidden layers. As the quantity of hidden layers within a neural layer increase, the model's feature learning becomes richer, and the model's prediction impact improves. Compared to the SNN, the DNN has a better learning capability and the method is illustrated in Fig. 2. This architecture allows the model to capture interactions between multiple student features, enabling prediction of personalized learning outcomes. Weight matrices are initialized using Xavier initialization, and dropout is applied to prevent overfitting.

DNN model training incorporates both upward and downward propagation. When using a DNN approach with two hidden layers and a “tanh” activation operation, the input layer represents $h(w)$, $h(w) \in Q^{m_1 \times 1}$. According to this calculation, m denotes the term vector size, Q represents the input feature space, and signifies data categories, indicating the initial hidden layer of the forward propagation includes m_1 neurons, w denotes the input vector, $h(w)$ represents input layer's activation function Eqs. (2 and 3).

$$y_j^{(1)} = G_1 h(w) + a_1 \quad (2)$$

$$y_p^{(1)} = \tanh(y_j^{(1)}) \quad (3)$$

The 2nd hidden layer has m_2 neurons, and its input and output are defined by Eqs. (4 and 5):

$$y_j^{(2)} = G_2 y_p^{(1)} + a_2 \quad (4)$$

$$y_p^{(2)} = \tanh(y_j^{(2)}) \quad (5)$$

$y_j^{(1)}$ and $y_j^{(2)}$ are the pre-activation output of hidden layers 1 and 2. $y_p^{(1)}$ and $y_p^{(2)}$ are post-activation output of hidden layers 1 and 2. The model's output layer (z) is represented in Eq. (6).

$$z = V \tanh(y_p^{(2)}) + a_3 \quad (6)$$

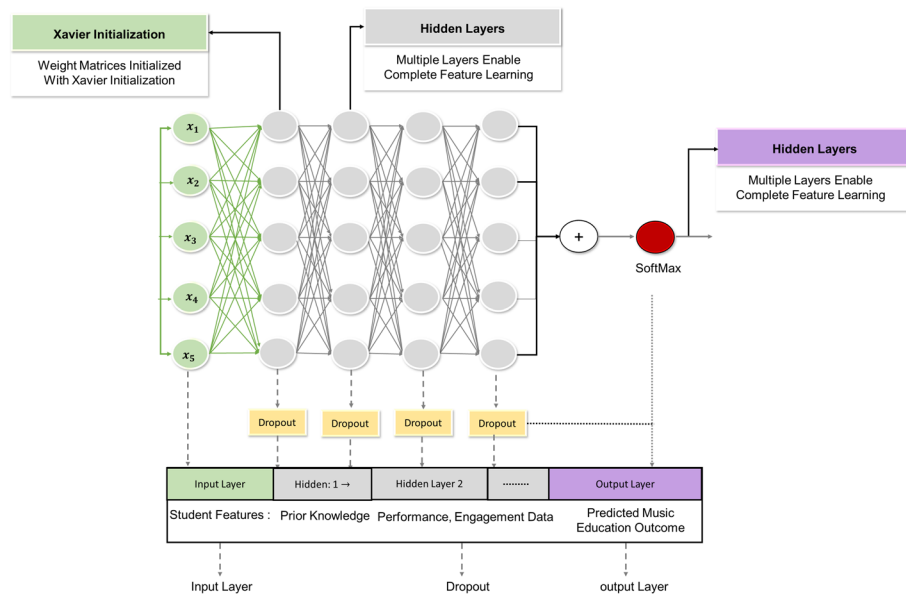


Fig. 2 Architecture of the DNN model

In the above Eqs. (2)-(6), G_1, G_2, V are Weight matrix structures for hidden layer one, hidden layer two, and output layer, whereas a_1, a_2, a_3 are threshold matrix for hidden layer one, hidden layer two, and output layer.

Employing θ to represent all the variables in the DNN, it becomes (Eq. (7)):

$$\theta = (G_1, G_2, V, a_1, a_2, a_3) \quad (7)$$

Apply $z \in Q^{m_3 \times 1}$ to demonstrate the operation of θ , z_j represents the log probabilities of sample j from the output layer; after regularization that is represented in the following Eq. (8).

$$o(e_j | w, \theta) = \frac{f^{z_j}}{\sum_{i=1}^{M_3} f^{z_i}} \quad (8)$$

Where e_j represents actual category label of training sample j , M denotes total number of training samples, o indicates SoftMax probability and f signifies SoftMax function. The variable $e^{(j)}$ represents the category of training sample j .

Backward propagation refers to the labelled samples $(w^{(j)}, e^{(j)})$, $j \in M$ is utilized as the total number of training samples, while $w^{(i)}$ denotes the training samples j . Discovering the variable θ to optimize the log-likelihood probabilities of $Q(\theta)$ with a standard term, the probability operation is appeared in the following Eq. (9):

$$K(\theta) = \sum_{j=1}^M \log o(e^{(j)} | w^{(j)}, \theta) + Q(\theta) \quad (9)$$

Where $K(\theta)$ denotes log-likelihood objective function with regularization. The randomized gradients ascending approach is applied to learn θ , and backward propagation technique is employed to iterate. Then, upgrade the variables until the desired precision is attained. The iteration expression is stated in the following Eq. (10):

$$\theta \leftarrow \theta + \eta \frac{\partial \log o(e | w, \theta)}{\tau \theta} \quad (10)$$

The variable η represents the learning rate. $\tau \theta$ denotes the gradient of the objective function with respect to all DNN parameters (θ). Thus, the DNN model predicted personalized learning paths and moderate accuracy, enabled initial skill and engagement improvements, but was less effective.

- Enriched Deep Neural Networks (EDNN)

The standard DNN model, while powerful in feature learning, often suffers from slower training due to numerous variables. To enhance its capacity for capturing complex student learning patterns, the model incorporates feature enrichment and deeper layers. The number of variables is linked to the choice of activation function, which also affects handling data imbalance. Common functions include sigmoid (Eq. 11), $\tanh$ (Eqs. 12–13), and ReLU (Eq. 14). ReLU is preferred for its ability to mitigate gradient vanishing issues, improving training efficiency and model performance in personalized learning path prediction. Feature enrichment combines prior knowledge, engagement metrics, and performance history, providing the model with richer contextual information. Hyperparameters such as the number of hidden units (16–128) and dropout rate (0.2–0.3) were optimized to ensure robust learning across diverse learners.

$$\sigma_t = \frac{1}{1 + e^{-t}} \quad (11)$$

$$\tanh = \frac{e^w - e^{-w}}{e^w + e^{-w}} \quad (12)$$

$$\tanh(w) = \frac{2}{1 + e^{-2w}} - 1 \quad (13)$$

$$\text{Relu} = \max(0, w) \quad (14)$$

Where σ_t denotes sigmoid role, $\tanh$ describes Hyperbolic Tangent Function, and t is neuron's input. Hence, the EDNN improved prediction accuracy and captured intricate learning behaviors, generating more personalized learning paths than the standard DNN, aiding personalized skill improvement.

- Attention mechanism (ATT)

ATT identifies and prioritizes the critical student features such as pitch accuracy, rhythm stability, and practice engagement, ensuring recommendations focus on impactful learning rates. Concentrated attention focuses on a single component in the input that is selected by maximum or random sampling technique, and it needs additional training to attain exceptional results.

Multi-head attention (2–4 heads) enables simultaneous focus on different aspects of performance, such as technical skills and engagement metrics. Attention weights are trained jointly with the EDNN to dynamically adapt to individual learner characteristics. However, soft attention is a procedure that allocates weights to all data to enable more efficient data usage. The soft attention mechanism calculates the attention score at moment (f_s) by applying a weight matrix X_b and b bias factor a , which acts on input components $w_1, w_2, \dots, w_S$, Eq. (15):

$$f_s = \tanh(X_b \{w_1, w_2, \dots, w_S\} + a) \quad (15)$$

Such scores are subsequently regularized by utilizing the SoftMax operation to generate the attention weights (b_s), which is represented in Eq. (16).

$$b_s = \frac{\exp(f_s)}{\sum_{l=1}^S \exp(f_l)} \quad (16)$$

Where f_l denotes attention score of the l -th input component. The attention mechanism is often separated into two phases: the first is to compute the attention structure, and the second is to calculate the weighted average of the input information using the attention population as a reference. The procedure starts with the attention scoring operation T . The outcome is sent to the SoftMax layer, which generates attention weights $(1, 2, \dots, m)$. The SoftMax layer is then supplied into the attention weights 1, 2, and so on; the attention weight vectors are weighted and averaged against the input data to produce the final result. Figure 3 depicts the attention method.

Therefore, this process highlights the key performance indicators, improving path prediction precision, accelerating skill mastery, and optimizing the personalized learning experience for each student.

2.3.2 Artificial lizard search optimization (ALSO)

The purpose of ALSO is to optimize the learning path parameters, continuously adjust the sequence and difficulty of tasks based on student progress, engagement, and feedback, and enhance the performance of the EDNN-ATT model. Population size (40–60 lizards) and maximum iterations (80–120) are tuned to ensure comprehensive search without excessive computation. Exploration probability (0.6–0.8) balances between global exploration and local refinement of personalized learning paths.

During this approach, lizards indicate the agents, and prey symbolizes the best answer. $W_j(s)$ denotes the location of the j th lizard at the s stage. Equation (17) calculates the subsequent location of the j th lizard.

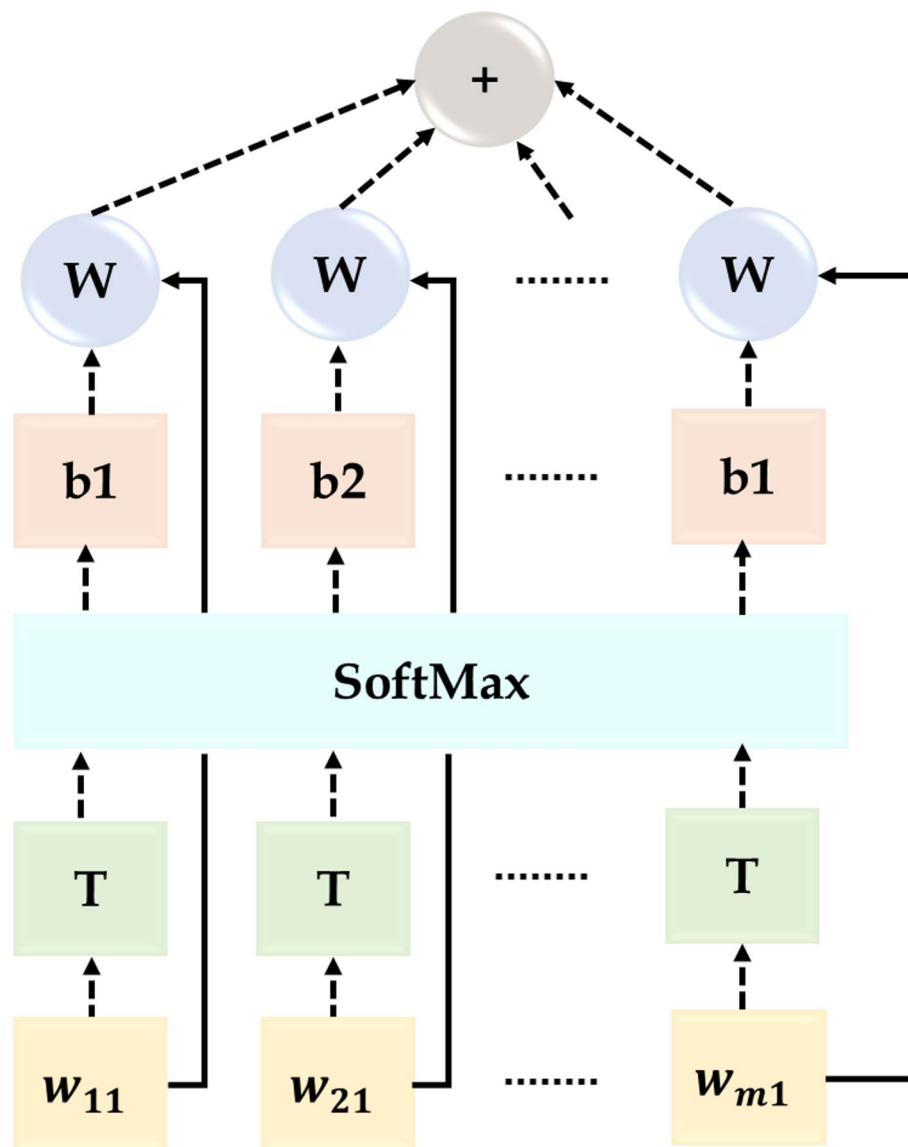


Fig. 3 Fundamental of the attention model

$$\begin{aligned}
 U1 &= d_1 q_1 (\text{obest} - W_j(s)) \\
 U2 &= d_2 q_2 (\text{hbest} - W_j(s)) \\
 W_j(s+1) &= W_j(s) + 0.3 \times \tau(s+1) \Delta\theta_j + U1 + U2
 \end{aligned} \tag{17}$$

The variables established for self-recognition and social interactions are d_1 , and d_2 . For the j -th lizard, $\Delta\theta$ denotes the distinction across the angle of the head and the tail in radians. q_1 , and q_2 represent random values between 0 and 1. $U1$ and $U2$ are movement contributions from self-recognition; obest indicates the present local optimal solution, whereas hbest represents the present global optimal solution. $W_j(s+1)$ defines as a randomly created number from 0 to q . The significant aspect for leaping movement is τ , Eq. (18):

$$r = \begin{cases} \tau_j(s) W_j = \text{obest} \\ \frac{E(\text{hbest}) - E(w_j)}{E(\text{hbest}) - E(\text{hworst})} \text{ else} \end{cases} \tag{18}$$

Where r denotes the movement factor controlling step size, $\tau_j(s)$ represents the random number controlling stochastic movement at stage s , $E(\text{hbest})$ denotes fitness of the global best solution, $E(\text{hworst})$ represents the fitness of the worst solution. $\Delta\theta$ is computed by Eq. (19).

$$\Delta\theta_j = (\theta_a - \theta_s) \times \frac{\pi}{180} \tag{19}$$

θ_a and θ_s represent the body angle and tail angle, bounded from -45 to 45 degrees. To compute θ_a and θ_s , directing body angles are to be computed. The directive body angles (ω_a) and tail angles (ω_s) are bounded from -10 to 10 . ω_s and ω_a are computed by Eq. (20) and the set of Equations (21).

$$S = \left[\omega_a(s), \frac{\omega_a^2(s) - a}{c0f}, \omega_s, \frac{-\omega_s^2(s) - d}{c - f} \right] + \left[0, \frac{-e}{c - f}, 0, \frac{-h}{c - f} \right] \times t_j(s) \tag{20}$$

$$\begin{aligned}
 b &= \frac{1}{2} k_a^2 k_s^2 n_a^2 n_s^2 \sin(2(\theta_a(s) - \theta_s(s))) \\
 J_a &= n_a k_a^2 J_s = n_s k_s^2 \\
 a &= (k_s^2 n_a n_s + J_s (n_a + n_s)) k_a k_s n_a n_s \theta_s^2(s) \sin(\theta_a(s) - \theta_s(s)) \\
 d &= (k_a^2 n_a n_s + J_a (n_a + n_s)) k_a k_s n_a n_s \theta_s^2(s) \sin(\theta_a(s) - \theta_s(s)) \\
 c &= (k_a^2 k_s^2 n_a^2 n_s^2 \cos(\theta_a(s) - \theta_s(s)))^2 \\
 f &= (k_a^2 n_a n_s + J_a (n_a + n_s)) (k_s^2 n_a n_s + J_s (n_a + n_s)) \\
 e &= (k_s^2 n_a n_s + J_s (n_a + n_s)) 0 k_a k_s n_a n_s \cos(\theta_a(s) - \theta_s(s)) (n_a + n_s) \\
 h &= (k_a^2 n_a n_s + J_a (n_a + n_s)) 0 k_a k_s n_a n_s \cos(\theta_a(s) - \theta_s(s)) (n_a + n_s)
 \end{aligned} \tag{21}$$

For j -th lizard, k_a represents body length, k_s denotes tail length, n_a indicates body mass, and n_s denotes tail mass. Meanwhile, these typical computations potentially alter for larger lizards. The lizard's complete length is denoted as k , and its total mass is n . a , b , c , d , e , f , h , J_a , J_s denote intermediate variables for computing angular adjustments and motion dynamics.

The proposed integration of ALS-EDNN-ATT offers a synergistic advantage by combining accurate prediction of student learning behaviors with adaptive optimization of personalized learning paths. The ALS-EDNN-ATT algorithm (Algorithm 1) generates highly efficient, adaptive learning paths, enhancing skill mastery, engagement, and task completion speed. By integrat-

ing accurate learning-behavior prediction with personalized path optimization, the model delivers dynamic, responsive music education. Hyperparameters are detailed in Table 3

Algorithm 1 ALS-EDNN-ATT model for personalized learning path planning and optimization model for music education

Start

Input:

- Student data $W = \{\text{prior knowledge, performance accuracy, engagement metrics}\}$
- Training labels $E = \{\text{learning outcomes categories}\}$
- Learning rate η
- Maximum epochs MaxEpochs

Initialize:

- Initialize EDNN weights $G1, G2, \text{ and biases } a1, a2, a3$ (Equation 7)
- Initialize Attention weights X and bias a (Equation 15)
- Initialize ALS parameters: lizard positions $W_j(s)$, body/tail angles θ_a, θ_s , stepsizes τ (Equations 17-21)
- Set iteration counter epoch = 0

EDNNModule:

For each sample w in W :

Forward Propagation

$$y_j^{(1)} = G_1 h(w) + a_1$$

$$y_p^{(1)} = \tanh(y_j^{(1)})$$

$$y_j^{(2)} = G_2 y_p^{(1)} + a_2$$

$$y_p^{(2)} = \tanh(y_j^{(2)})$$

$$z = V \tanh(y_p^{(2)}) + a_3$$

$$o(e_j | w, \theta) = \frac{f_j}{\sum_{i=1}^M f_i}$$

Backward Propagation

$$\theta \leftarrow \theta + \eta \frac{\partial \log o(e | w, \theta)}{\partial \theta}$$

EDNN with ATTModule:

For input components $w_1, w_2, \dots, w_S$:

$$f_s = \tanh(X_s \{w_1, w_2, \dots, w_S\} + a)$$

$$b_s = \frac{\exp(f_s)}{\sum_{j=1}^S \exp(f_j)}$$

Weighted average for output

$$\text{weighted_input} = \sum_{s=1}^S \{s=1\}^S (b_s * w_s)$$

ALSModule:

For each lizard at stages:

Compute movement contributions

$$U1 = d_1 q_1 (\text{obest} - W_j(s))$$

$$U2 = d_2 q_2 (hbest - W_j(s))$$

$$W_j(s+1) = W_j(s) + 0.3 * \tau(s+1) \Delta \theta_j + U1 + U2$$

Update step using if-else (Equations 18)

If w_j meets sobest criteria:

$$\tau_j(s+1) = \text{obest}$$

Else:

$$\tau_j(s+1) = (E(hbest) - E(w_j)) / (E(hbest) - E(hworst))$$

Compute angled differences $\Delta \theta_j$ (Equations 19-21)

$$\Delta \theta_j = (\theta_a - \theta_s) * \pi / 180$$

ALS-EDNN-ATT Integration:

While epoch < MaxEpochs:

1. Train EDNN on student data (forward + backward propagation)
2. Apply ATT to emphasize key features
3. Optimize learning paths with ALS based on predicted performance
4. Update global and local best paths (hbest, obest)
5. epoch = epoch + 1

Output:

- Personalized Learning Paths
- Predicted performance improvements
- Engagement metrics

End

2.4 Personalized learning path generation

The ALS-EDNN-ATT model personalizes music education by generating individualized learning paths, adjusting module difficulty, practice frequency, and feedback based on each learner's skills and engagement. Framed as a multiclass classification, it predicts experience levels for vocal (novice/intermediate/advanced) and instrumental learners. ALS structures learning steps, EDNN captures non-linear relationships among knowledge and performance, and ATT weights domain-specific features, enabling optimized

Table 3 Hyperparameters of the proposed ALS-EDNN-ATT model

Hyperparameter	Value / Range
Hidden Units (EDNN)	16–128
Number of Layers (EDNN)	4
Learning Rate	0.001–0.005
Batch Size	32–64
Activation Function	ReLu
Dropout Rate	0.2–0.3
Optimizer	Adam
Epochs	100
Attention Heads	2–4
Population Size (ALSO)	40–60
Maximum Iterations (ALSO)	80–120
Exploration Probability (ALSO)	0.6–0.8

recommendations that enhance learning efficiency and skill development. The integration of EDNN predictions, ATT weighting, and ALSO optimization ensures that generated paths reflect both current learner performance and adaptive progression strategies. Paths are continuously refined to maximize skill acquisition efficiency, engagement, and knowledge retention.

The experience-level categorization adopted in the research is grounded in established pedagogical practices used in formal music education rather than being arbitrarily rule-based. For vocal learners, theory pre-test score thresholds (< 50, 50–75, > 75) align with commonly employed grading rubrics in structured music curricula, where scores below 50 indicate insufficient foundational understanding, scores between 50 and 75 reflect developing competency, and scores above 75 represent advanced conceptual and practical mastery. Similarly, for instrumental learners, practice-duration thresholds (< 3 years, 3–5 years, > 5 years) correspond to widely recognized stages of instrumental skill acquisition, motor coordination development, and repertoire complexity progression in formal training environments.

2.5 Hypothesis development

To ensure a falsification-based empirical framework, the research formulates the following hypotheses aligned with the research objective of personalized music education using the ALS-EDNN-ATT model:

H1 The ALS-EDNN-ATT model’s predictions of student final performance are significantly associated with prior knowledge, engagement, and skill-related variables.

H2 Adaptive adjustments in learning module sequencing, task difficulty, and practice frequency by the ALS component significantly improve students’ skill progression and overall performance compared to non-adaptive learning paths.

Hypotheses were tested using quantitative metrics including predicted final performance, learning progression rates, and engagement indicators such as weekly practice hours, lesson completions, and discussion participation. One-way and two-way ANOVA analyses were applied to evaluate (H1) the significance of associations between ALS-EDNN-ATT predictions and prior knowledge, engagement, and skill-related variables, and (H2) the impact of adaptive learning path adjustments on students’ skill progression and overall performance across vocal and instrumental learners at novice, intermediate,

and advanced levels. Hyperparameters such as learning rate (0.001–0.005), batch size (32–64), and attention heads (2–4) are carefully selected to ensure reproducibility and stability of hypothesis testing. The hypotheses are empirically validated using one-way and two-way ANOVA analyses, ensuring statistical rigor.

3 Performance analysis

The DL-based personalized learning path planning and optimization model for music education was established effectively. The experiments are conducted on Python 3.11 with TensorFlow, PyTorch, NumPy, Pandas, Scikit-learn, an Intel Core i9-12900 K processor, an NVIDIA RTX 3090 GPU, 64 GB RAM, and Windows 11 Pro, SPSS 27.0 for hypothesis model evaluation using statistical ANOVA test. This section involves the experimental findings, student learning outcomes and efficiency, student progression trends over 12 weeks, assessment of engagement metrics, performance analysis by learning style, qualitative feedback from teachers and students, and comparative analysis. While the ANOVA analysis provides insights into the significance of various factors affecting the model's predictions, additional model validation techniques are essential for ensuring the robustness and generalizability of the proposed deep learning and optimization framework. In this study, incorporate cross-validation to evaluate the model's performance on different subsets of the data, ensuring that the model's predictions are not overfitting to the training data. The research also performs benchmarking against established baseline models to provide a comparative performance evaluation, further enhancing the methodological rigor of the research.

3.1 Experimental assessment

Figure 4 illustrates the positioning of students based on theory pretest, ear training, and harmony scores, with scale rhythm performance indicated by color. This visualization identifies correlations among foundational musical skills and highlights differences between vocal and instrumental learners. In the ALS-EDNN-ATT model, these features serve as input vectors for the EDNN, enabling the network to predict performance outcomes. The observed patterns inform the ATT, which prioritizes features such as weak rhythm skills for adaptive practice. Additionally, the distribution helps the ALSO optimization module determine task sequencing and difficulty for students with varying skill levels.

Figure 5 presents overlapping performance curves for instrumental features, showing variability across pitch, rhythm, and technical accuracy. These variations reflect the complexity of student learning behaviors, which the EDNN captures through non-linear transformations. The attention mechanism uses this information to assign higher weights to critical features, ensuring the model emphasizes areas requiring intervention. The ALSO module leverages these patterns to adjust practice focus, task difficulty, and sequencing, targeting areas with high variability to accelerate skill acquisition.

Figure 6 displays multidimensional student performance across theory, ear training, and scale rhythm metrics, revealing clusters, correlations, and outliers. The EDNN uses these inputs to model expected outcomes, while ATT emphasizes the features with the most predictive value. The clusters identified also guide the ALSO module in determining optimal learning paths: students with similar performance trends may receive

analogous sequences of tasks, while outliers are assigned tailored interventions to address specific weaknesses.

Figure 7 shows hierarchical clustering of students based on performance, engagement, and learning preferences, highlighting natural groupings. This clustering informs the EDNN about latent patterns in the input data and supports ATT in assigning feature importance within each group. The ALS-EDNN-ATT model uses these clusters to adapt learning paths dynamically, ensuring module sequencing and difficulty adjustments align with each student's cluster-based learning profile.

Figure 8 presents the distribution of theory pretest scores, combining raw frequency and density estimates. EDNN uses these scores as numerical features to predict future performance, while ATT prioritizes features representing weak or outlying scores. Students with lower scores are guided by ALSO to begin with simpler modules, while high-performing students progress to advanced tasks. This figure links directly to adaptive difficulty assignment in personalized paths.

Figure 9 compares theory pretest and ear training scores for individual students, highlighting performance alignment and discrepancies. The EDNN captures these non-linear relationships, and ATT emphasizes weaker skill domains. ALS uses these insights to customize practice recommendations, ensuring students strengthen deficient areas while maintaining overall balance in skill development.

Figure 10 visualizes correlations among 21 musical training, performance, and engagement variables. Positive and negative dependencies reveal critical relationships such as the effect of practice consistency on overall performance. EDNN incorporates these correlations in its prediction model, ATT assigns higher attention to impactful variables, and ALS optimizes learning paths by adjusting module priority and practice frequency based on these interdependencies.

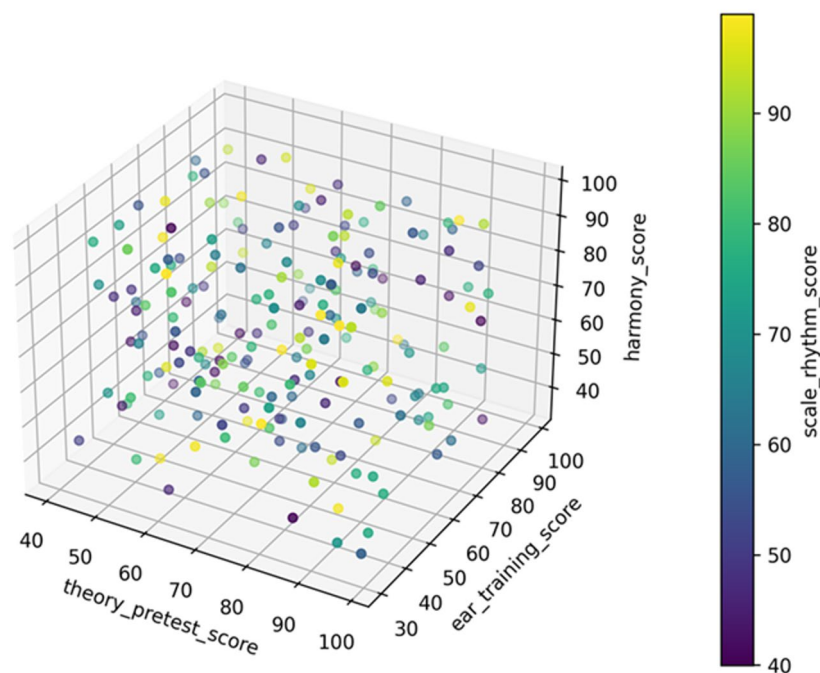


Fig. 4 Musical training performance mapping

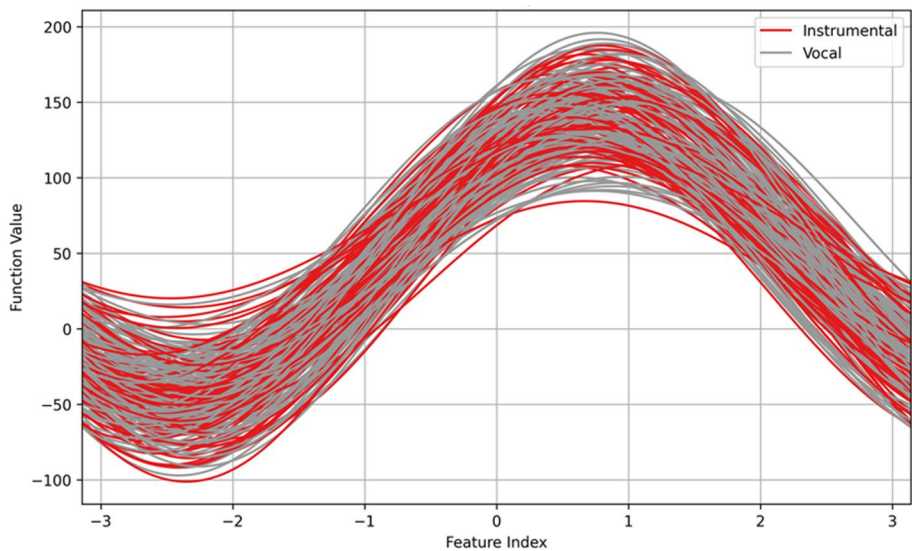


Fig. 5 Instrumental and vocal feature dynamics

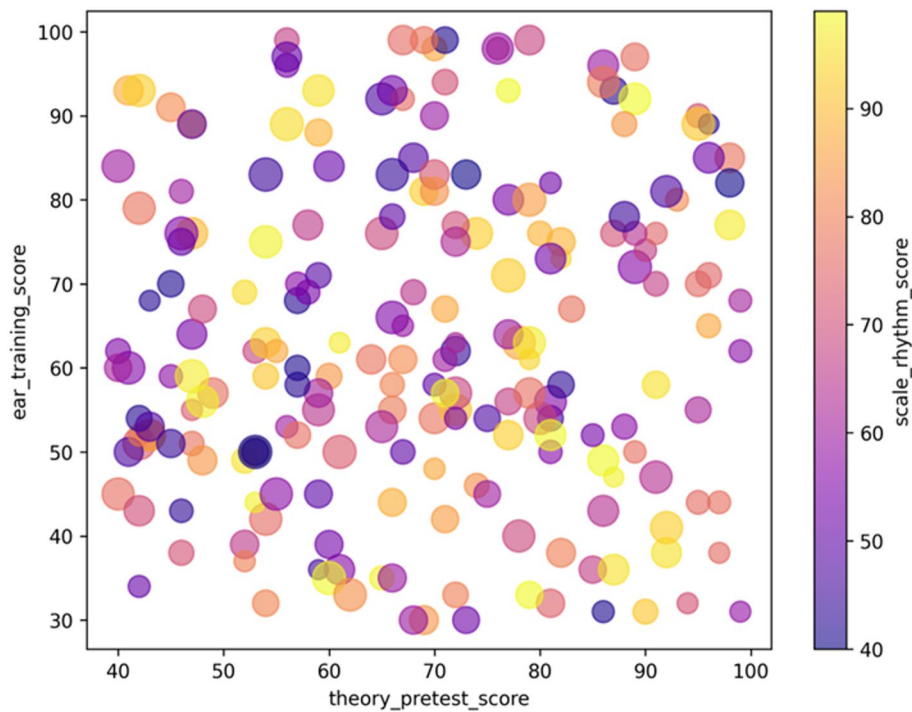


Fig. 6 Music student performance distribution overview

Figure 11 tracks student progression across consecutive time points, identifying improvements, declines, and consistency. EDNN predicts expected learning trajectories, while ATT prioritizes features showing stagnation or decline. ALS leverages this temporal information to dynamically modify the next learning step, ensuring continued skill growth and engagement.

Figure 12 highlights individual score variations, clusters, and outliers among 30 students. EDNN uses these distributions to model expected performance outcomes. ATT emphasizes critical areas of weakness, while ALS ensures that module sequencing, task

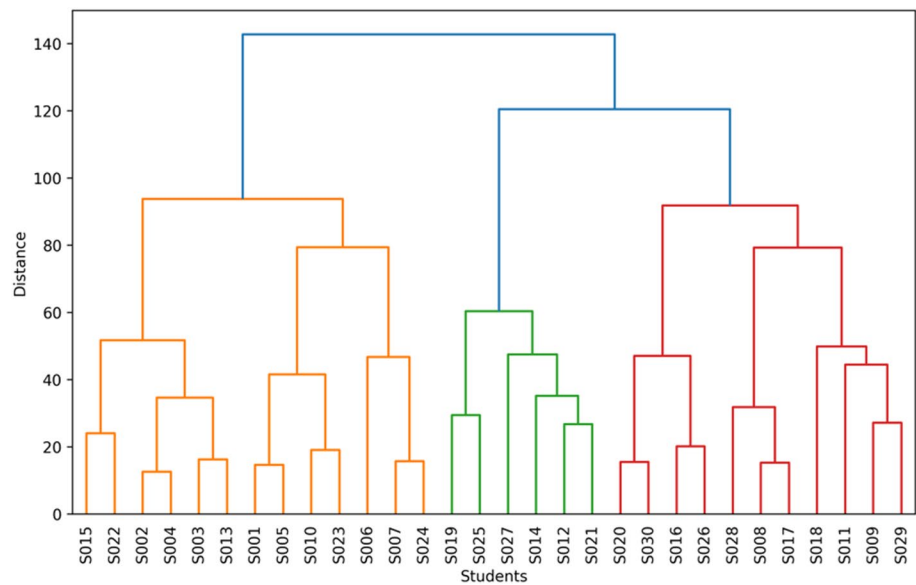


Fig. 7 Student performance cluster analysis overview

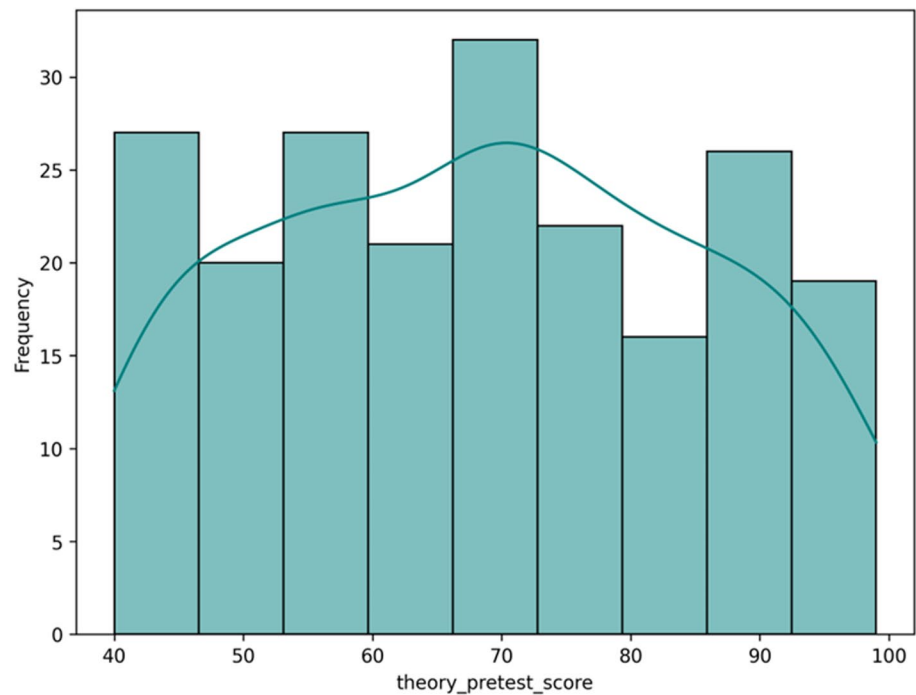


Fig. 8 Theory pretest score distribution overview

difficulty, and practice frequency are personalized to address student-specific learning needs, facilitating accelerated skill acquisition.

Figure 13 presents pairwise relationships among theory, ear training, harmony, and scale-rhythm scores. EDNN models these non-linear interactions, ATT prioritizes features with the strongest predictive influence, and ALS translates these insights into tailored module sequencing. This allows students to focus on interrelated skills that require simultaneous development for holistic musical proficiency.

Figure 14 differentiates performance patterns across vocal and instrumental tracks, revealing individual consistency and disparities. EDNN predicts outcomes considering track-specific characteristics, ATT assigns domain-specific weights to features such as pitch accuracy or motor skill complexity, and ALS sequences learning modules and adjusts task difficulty to align with each track’s unique learning profile.

Figure 15 highlights predictive relationships and slight positive correlations between theory and ear training scores. EDNN captures this relationship, ATT emphasizes weaker domains (e.g., auditory skills lagging behind theory), and ALS recommends targeted practice exercises to balance skill development, ensuring personalized interventions.

Figure 16 presents temporal trends of theory pretest scores with a smoothed rolling mean. EDNN uses this trend to forecast progression, ATT weights critical deviations, and ALS adjusts module difficulty and sequencing over time. This figure demonstrates how temporal learning data informs dynamic adaptation of personalized paths.

Figure 17 visualizes student progression across novice, intermediate, and advanced levels for vocal and instrumental tracks. EDNN predicts transitions between levels, ATT highlights features driving progression, and ALS sequences the next set of learning modules to maximize advancement along personalized trajectories.

Figure 18 presents cumulative contributions of theory, ear training, and harmony scores. EDNN predicts expected cumulative outcomes, ATT assigns higher weights to dominant or lagging metrics, and ALS optimizes learning paths to focus practice and sequencing on weaker skill domains, ensuring balanced skill development.

Figure 19 compares theory pretest performance between vocal and instrumental students, revealing differences in score distributions and central tendencies. EDNN captures these track-specific characteristics, ATT emphasizes features driving gaps, and ALS customizes module sequencing and difficulty adjustments to align with each track’s distinct learning needs.

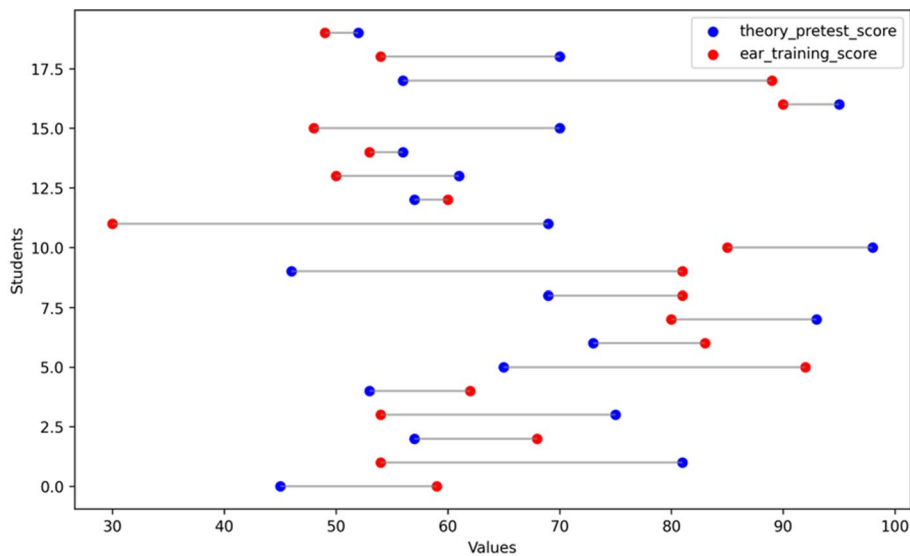


Fig. 9 Individual music performance comparison chart

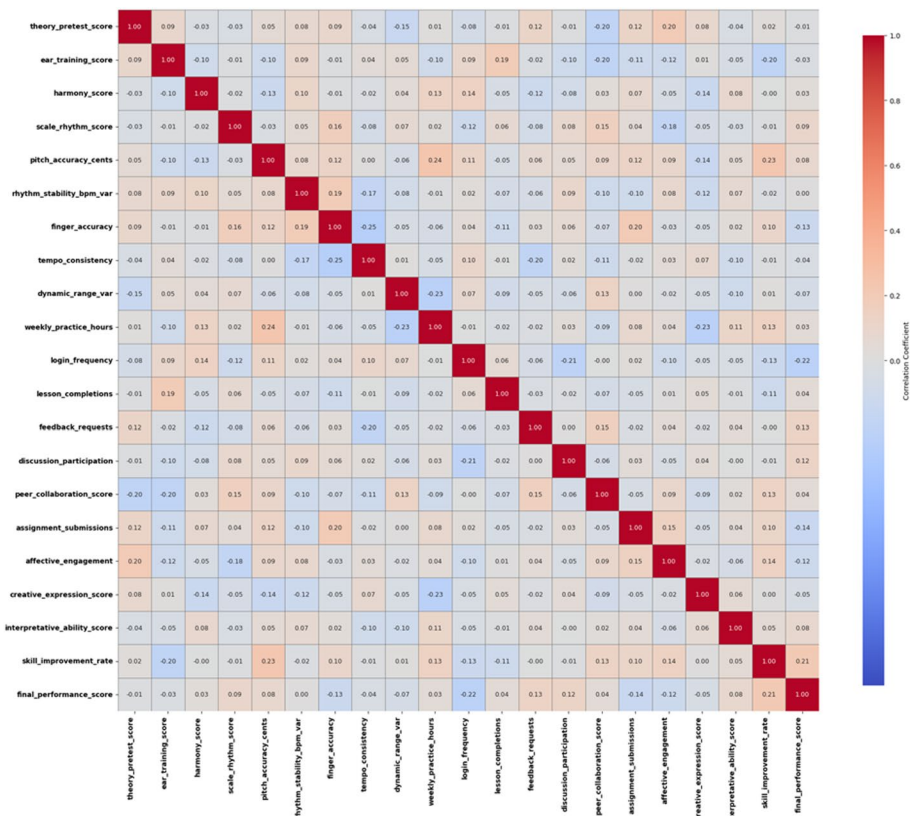


Fig. 10 Music student variable correlation map

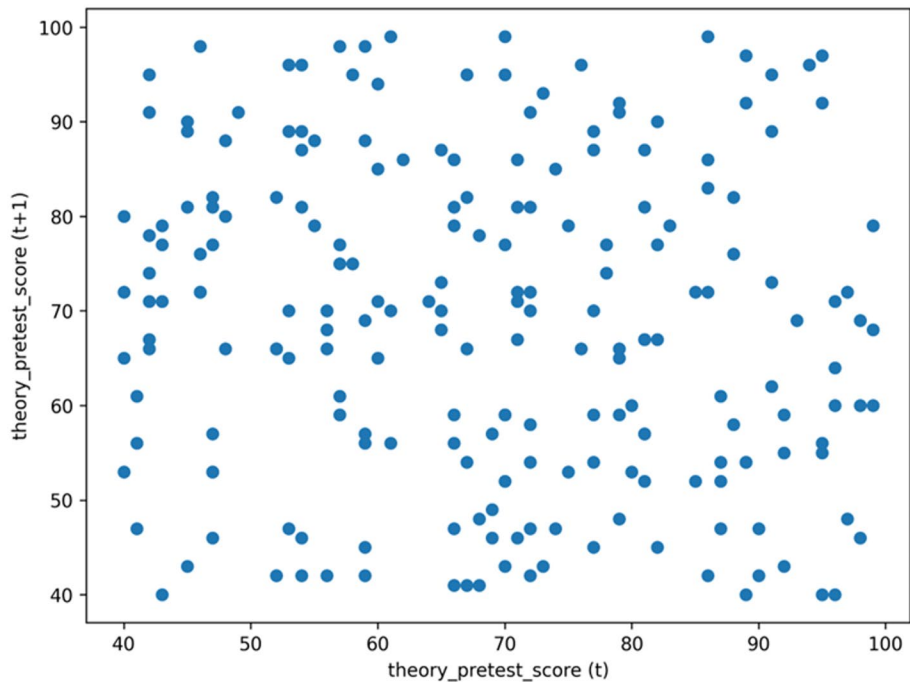


Fig. 11 Student score progression analysis

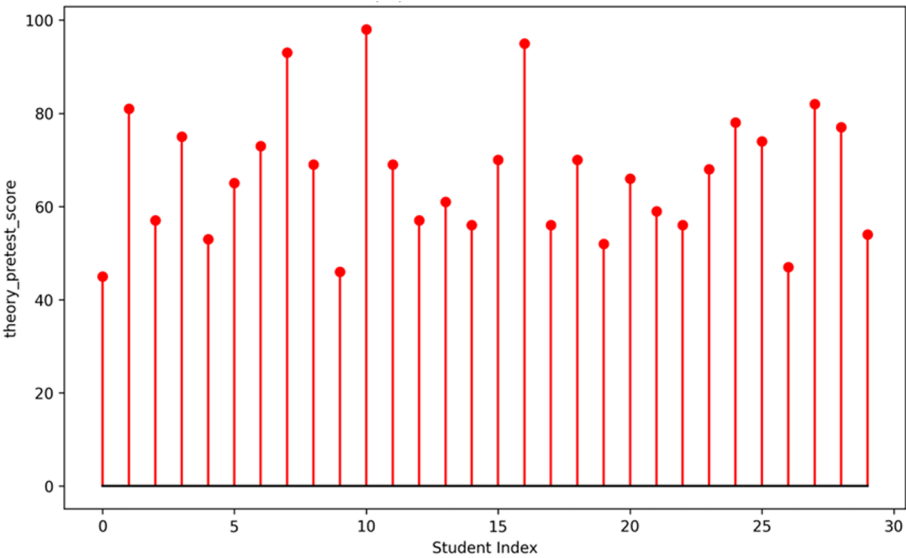


Fig. 12 Individual student score distribution

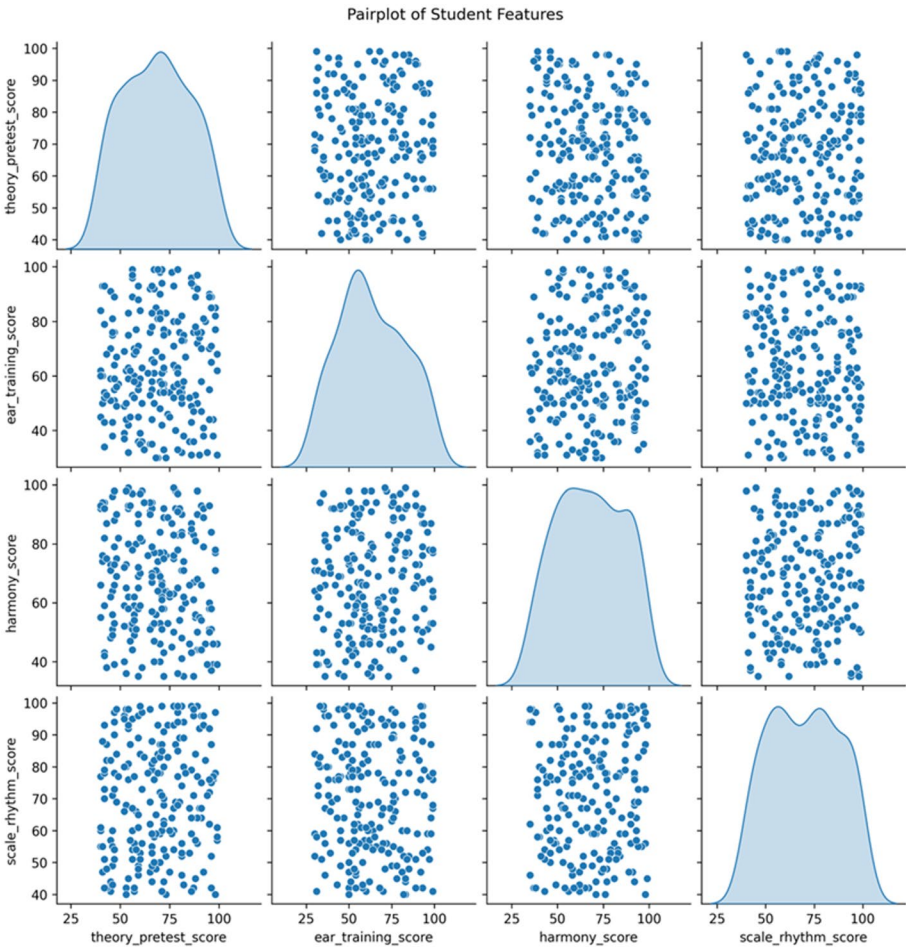


Fig. 13 Student performance metrics relationships

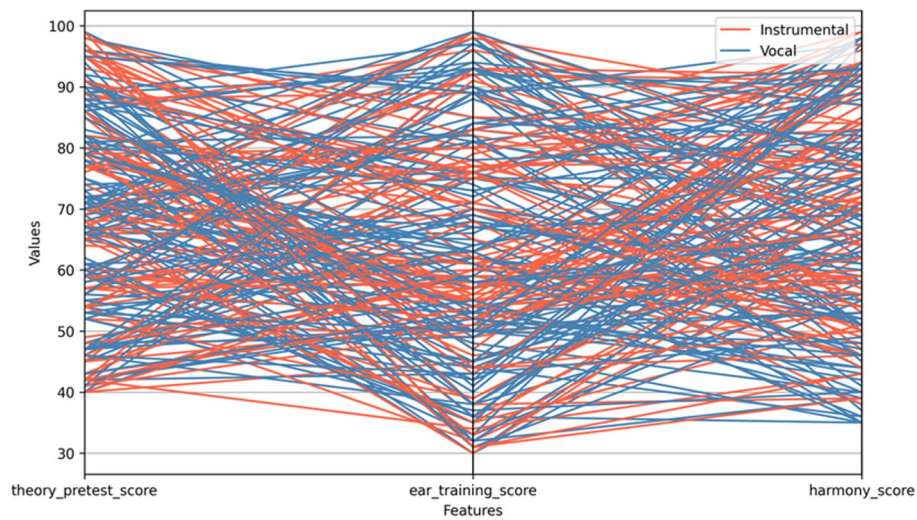


Fig. 14 Multidimensional student performance comparison

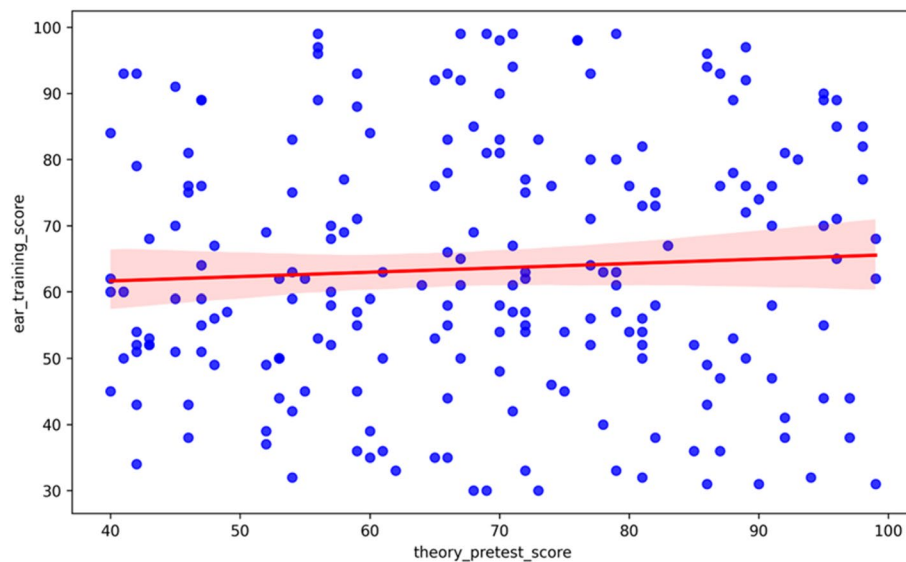


Fig. 15 Theory versus ear training

3.2 Student learning outcomes and efficiency

Figure 20 quantify student skill gains, mastery speed, and weekly progression over 12 weeks. EDNN predicts expected improvements, ATT emphasizes critical skill areas for attention, and ALS adapts learning paths dynamically based on progression trends. These figures demonstrate the model's ability to translate feature analysis into actionable, personalized recommendations, accelerating skill mastery, enhancing engagement, and improving efficiency.

This presents quantitative results of student skill development and learning efficiency before and after implementing the ALS-EDNN-ATT model. It combines accuracy, mastery, completion time, and attempt frequency to show the model's impact on accelerating skill acquisition and reducing redundant practice. Each metric highlights a key aspect of personalized learning: performance gains, efficiency improvements, and faster knowledge mastery. Therefore, students achieved substantial performance improvements,

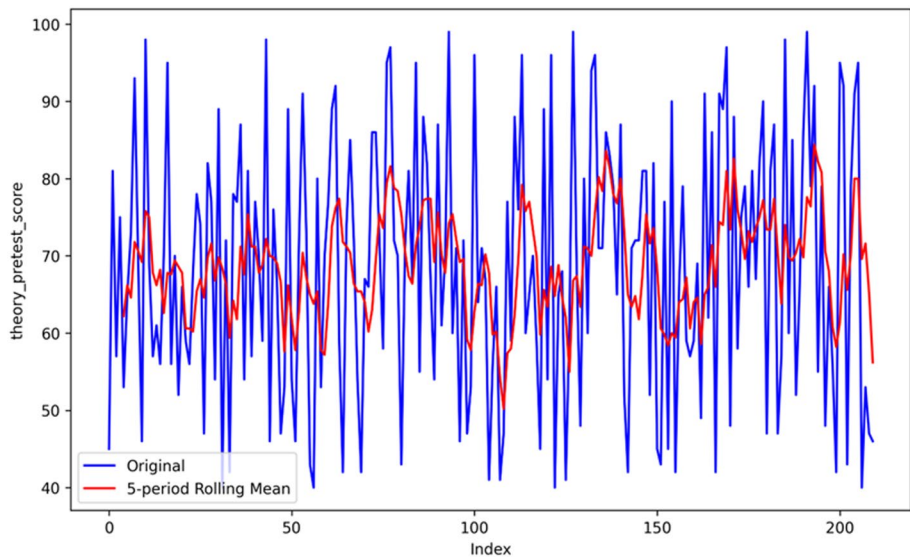


Fig. 16 Rolling trend theory scores

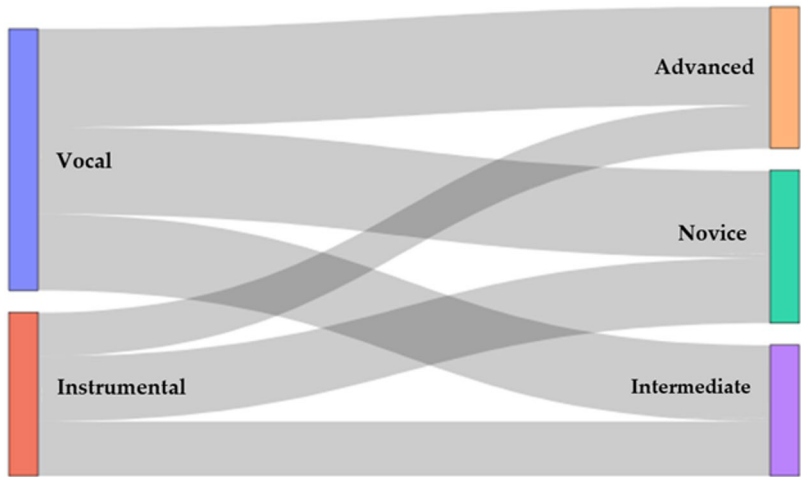


Fig. 17 Student track proficiency flow

faster task completion, and higher mastery rates, confirming that personalized learning paths significantly enhance skill development and learning efficiency.

3.3 Student progression trends over 12 weeks

Students’ weekly progression across musical skills is tracked to demonstrate incremental improvement and the effectiveness of personalized learning paths over a 12-week training period. Figure 21 shows the visual representation of student progression.

This details the weekly development of instrument performance, vocal stability, music theory comprehension, and creative expression for all students. It offers cumulative improvement and shows how the ALS-EDNN-ATT model supports continuous learning. The incremental increases illustrate the model’s capability to adapt learning paths to student progress, reinforcing both skill mastery and engagement over time. Thus, students showed steady skill improvement across all domains over 12 weeks, validating that adaptive learning paths progressively enhance musical proficiency and retention.

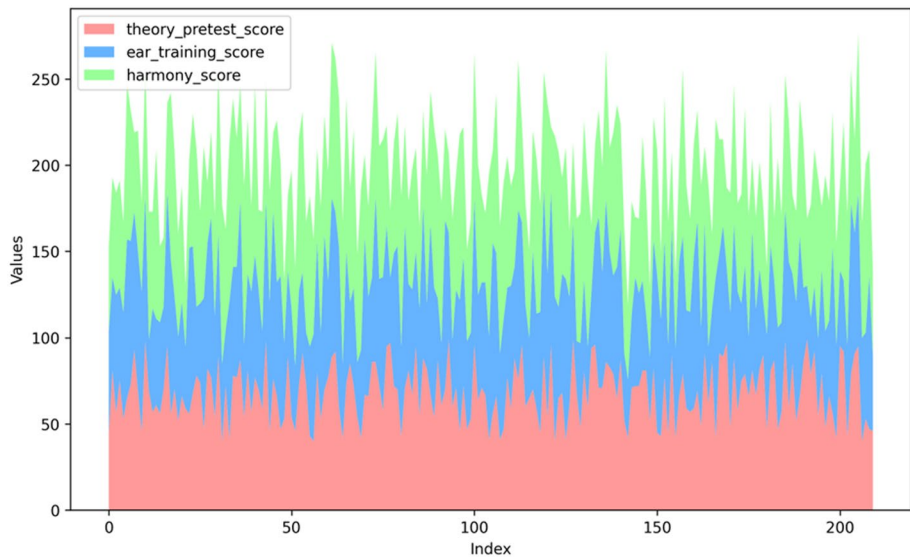


Fig. 18 Cumulative music performance metrics

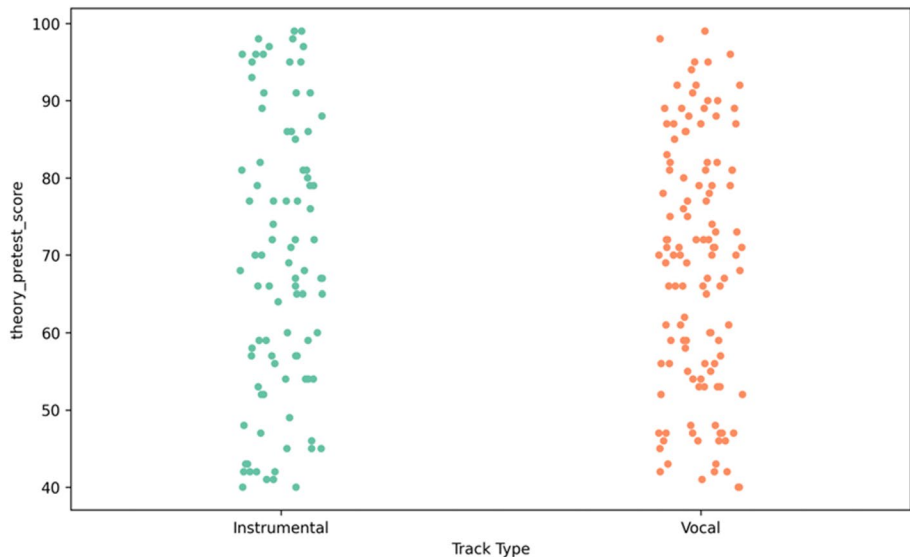


Fig. 19 Track-based theory score comparison

3.4 Assessment of engagement metrics

Table 4 shows the engagement metrics outcomes that measure student engagement improvements, including learning time, participation, and practice frequency, to assess the effectiveness of personalized learning paths in motivating music learners.

This quantifies the increase in student engagement after ALS-EDNN-ATT implementation. Metrics include average weekly learning hours, active participation rate, and number of practice sessions. The increases demonstrate that personalized, adaptive learning paths not only improve skills but also encourage more consistent, motivated practice and active involvement in music learning activities. Hence, the engagement metrics significantly increased, indicating that personalized learning paths motivate students, enhance participation, and promote sustained practice, supporting improved overall learning outcomes.

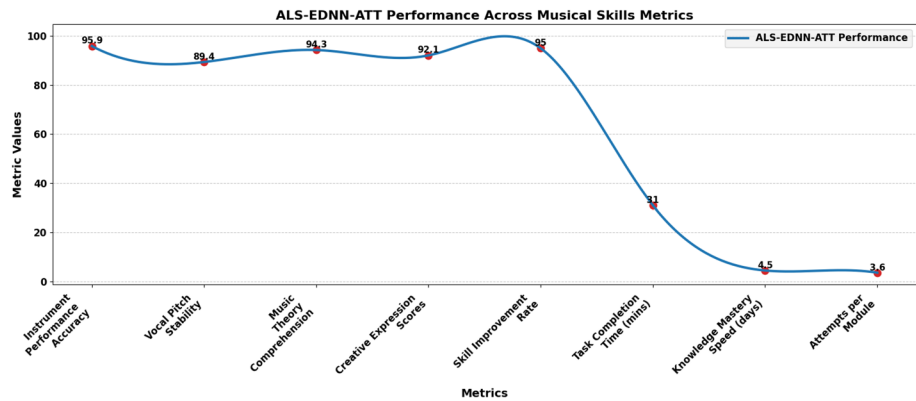


Fig. 20 Student learning outcomes and efficiency

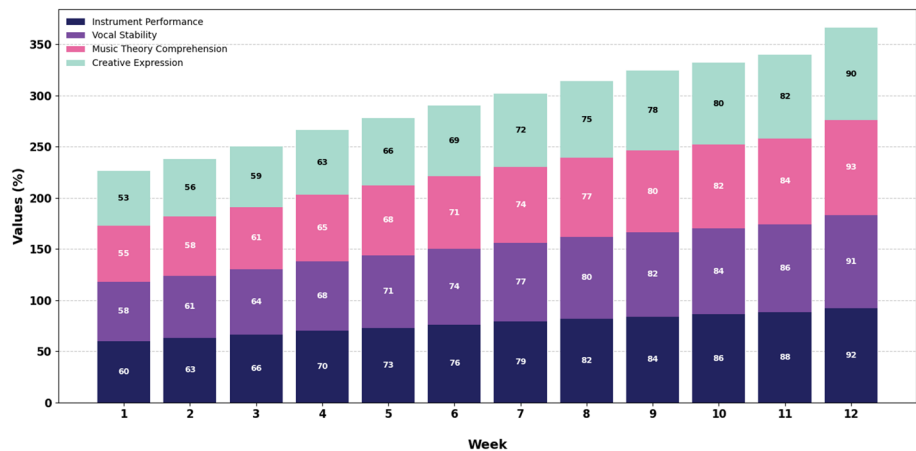


Fig. 21 Graphical illustration of student progression across musical skills

3.5 Performance analysis by learning style

Table 5 offers the learning style findings that analyze the effectiveness of personalized learning paths across different learning styles to demonstrate adaptability and differential impact on skill, engagement, and efficiency.

This presents outcomes segmented by learning style such as visual, auditory, and kinesthetic. It shows how the proposed ALS-EDNN-ATT accommodates individual preferences, with auditory learners benefiting slightly more. Differences across metrics confirm the model’s adaptability and its ability to tailor learning paths according to preferred modes of learning, ensuring optimal engagement and skill acquisition for all students. Therefore, the personalized learning paths effectively improve skills, engagement, and task efficiency for all learning styles, demonstrating the model’s adaptability and equitable impact on diverse student learners.

3.6 Qualitative feedback from teachers and students

The qualitative feedback captures subjective teacher and student perspectives on personalized learning paths to evaluate perceived adaptability, engagement, and overall satisfaction in music education, Table 6.

This summarizes qualitative feedback from both teachers and students, highlighting adaptability, instructional support, engagement, and perceived learning improvements.

Table 4 Engagement metrics

Metric	ALS-EDNN-ATT[Proposed]
Average Learning Time (hrs/week)	7
Active Participation (%)	96%
Practice Sessions per Week	6

Table 5 Performance efficiency by learning style

Learning Style	Average Skill Gain (%)	Engagement Increase (%)	Task Completion Time Reduction (%)
Visual	28.7	32.4	29.1
Auditory	31.2	35.1	32.5
Kinesthetic	29.5	33.7	30.4

Teachers reported enhanced efficiency and focus on creativity, while students valued clarity, relevance, and motivation. The ratings demonstrate strong acceptance and perceived effectiveness of the personalized learning system in complementing quantitative improvements in skills and efficiency. Thus, the feedback indicates high satisfaction and strong perceived improvements, supporting that personalized learning paths are effective, engaging, and well-received by both teachers and students.

3.7 Hypothesis testing result

The ANOVA-based hypothesis testing is employed to examine the statistical significance of learner attributes and adaptive learning path components on educational outcomes. However, ANOVA is not used as the sole validation mechanism; rather, it complements model-oriented evaluations by validating the educational impact of the ALS-EDNN-ATT predictions and optimization strategies. The ANOVA analysis evaluated six key factors influencing personalized music learning outcomes, including prior knowledge, engagement, skill metrics, module sequencing, task difficulty, and practice/feedback. All factors showed significant effects on student performance ($p \leq 0.003$), confirming that both learner attributes and adaptive interventions meaningfully impact final outcomes. These results validate the effectiveness of the ALS-EDNN-ATT model in optimizing personalized learning paths. This test outcome is demonstrated in Table 7.

3.8 Comparative analysis

This comparative evaluation serves as a model-oriented validation by demonstrating that the proposed ALS-EDNN-ATT framework not only achieves statistically significant outcomes but also consistently outperforms state-of-the-art deep learning models in predictive accuracy, robustness, and adaptability for personalized music learning. The comparative analysis compares the proposed method against existing methods by using performance metrics that ensure superior personalization, predictive accuracy, efficiency, and adaptability in personalized music education. The existing methods [30] include Softmax with Backpropagation (Softmax-BP), Improved Local Binary Patterns (ILBP), Audio Spectrogram Transformer (AST), Pretrained Audio Neural Networks (PANNs), Wav2Vec 2.0, Convolutional Recurrent Neural Network (CRNN), YAMNet, and Attention-based Gated Recurrent Convolutional Neural Network (A-GRCNN). The utilized performance metrics are accuracy, precision, recall, and f1-score. The existing

Table 6 Findings of qualitative feedback

Feedback Source	Aspect Evaluated	Observations	Rating (1–5)
Teachers	Adaptability to student needs	Personalized paths allowed targeted intervention and reduced repetitive drills	5
Teachers	Instructional support	Easier to monitor progress and focus on creative teaching	4.8
Teachers	Efficiency in teaching	Reduced time spent on manual adjustments	4.7
Students	Relevance of learning path	Learning modules aligned with personal pace and goals	4.9
Students	Engagement and motivation	Increased motivation due to interactive, adaptive recommendations	4.8
Students	Learning satisfaction	High satisfaction with clear progression and actionable feedback	4.9
Students	Perceived improvement	Felt skill development was faster and more effective	4.8

models utilize a distinct dataset, which, while relevant, lacks some of the specific features present in dataset, such as detailed domain-specific engagement metrics and prior knowledge. The research dataset, sourced from Kaggle, provides more nuanced data on student performance, practice behavior, and engagement, which better supports the personalized learning path optimization. This makes our comparison more robust and aligns with our model’s strengths in adapting to real-time learning progress and engagement.

- **Accuracy:** This measures overall correctness of classification, showing how well the model predicts music learning outcomes by balancing all correctly identified performance and engagement categories.
- **Precision:** It indicates the reliability of positive predictions, ensuring recommended learning paths truly match the student’s strengths without misleading or irrelevant adaptations, which is crucial for personalized music education.
- **Recall:** This evaluates completeness by checking how well the model identifies all relevant learning behaviors, ensuring no critical student skill or engagement pattern is missed.
- **F1-score:** It balances precision and recall, providing a single robust metric for model effectiveness in capturing both correct and comprehensive learning path recommendations. Table 8; Fig. 22 provide the findings of the proposed and existing methods using performance metrics.

Compared with multiple baselines, ALS-EDNN-ATT achieves the best performance, attaining 0.984 accuracy, 0.981 precision, 0.978 recall, and 0.978 F1-score, demonstrating superior reliability, comprehensive learning factor capture, and consistently effective personalized learning path adaptation.

Inference time per query (ms) Time taken by the model to make a prediction for a single input; lower values indicate faster real-time performance.

Number of parameters (Million, M) Total trainable weights in the model; smaller numbers often mean lighter, more memory-efficient models.

GPU memory usage (GB) Amount of GPU memory consumed during inference; lower usage allows deployment on devices with limited GPU resources.

Table 7 ANOVA Results for Hypothesis Testing of ALS-EDNN-ATT Model

Factors	Degree of freedom (df)	Sum of Squares (SS)	Mean Square (MS)	F-value	p-value	Interpretation
Prior Knowledge (theorypretestscore)	2, 207	120.5	60.25	15.42	< 0.001	Significant association with final performance
Engagement Metrics (weeklypracticehours)	2, 207	95.3	47.65	12.11	0.002	Engagement significantly influences performance
Skill-Related Variables (scalerhythmscore)	2, 207	88.6	44.3	11.34	0.001	Skill metrics significantly correlate with outcomes
Learning Module Sequencing (lessoncompletions)	2, 207	102.4	51.2	13.22	0.001	Adaptive sequencing improves skill progression
Task Difficulty Adjustment (tempo consistency)	2, 207	78.9	39.45	10.18	0.003	Adaptive task difficulty positively affects outcomes
Practice Frequency & Feedback (feedbackrequests)	2, 207	91.7	45.85	11.76	0.002	Adaptive practice & feedback significantly improves performance

CPU inference time (ms) Time taken for prediction when running on a CPU instead of GPU; important for non-GPU environments.

Training time per epoch (s) Time required to process the full training dataset once; shorter times indicate faster training.

Total time for 5-fold cross-validation (h) Overall time to perform model evaluation using 5-fold CV; includes training and validation, reflecting computational efficiency.

Table 9 highlights the computational efficiency of the proposed ALS-EDNN-ATT model for IPE resource recommendation. The model achieves a low inference time of 8 ms per query, enabling fast real-time recommendations. It maintains a compact architecture with only 0.3 million parameters, resulting in minimal GPU memory consumption (0.2 GB). The CPU inference time is limited to 55 ms, making the model suitable for deployment on resource-constrained systems. Additionally, the model demonstrates efficient learning with a training time of 2 s per epoch and completes 5-fold cross-validation in 0.3 h, confirming its overall scalability and resource efficiency.

3.9 5-Fold cross validation

The ALS-EDNN-ATT model exhibits high consistency across folds, with minimal standard deviation, confirming robustness and generalization. Average metrics closely match the single-run evaluation reported previously (Accuracy: 0.984, F1-score: 0.978), indicating low overfitting. Table 10 cross-validation demonstrates the model's stable performance across different partitions of the student dataset, enhancing confidence in the model's predictive reliability for personalized music learning paths.

4 Discussion

The research contributes to the literature by moving beyond performance prediction and demonstrating how DL and optimization can be jointly employed to realize actionable, adaptive learning path personalization in formal music education. The existing methods have some drawbacks, as follows: Softmax-BP [30] struggles with complex sequential

Table 8 Numerical comparative outcomes for proposed and existing methods

Methods	Accuracy	Precision	Recall	F1-score
Softmax-BP [31]	0.959	0.960	0.953	0.957
ILBP [31]	0.960	0.971	0.957	0.97
AST [31]	0.925	0.921	0.913	0.918
PANNs [31]	0.916	0.91	0.90	0.905
Wav2Vec 2.0 [31]	0.943	0.952	0.948	0.947
CRNN [31]	0.873	0.868	0.865	0.868
YAMNet [31]	0.868	0.860	0.847	0.855
A-GRCNN [31]	0.961	0.970	0.958	0.963
ALS-EDNN-ATT [Proposed]	0.984	0.981	0.978	0.978

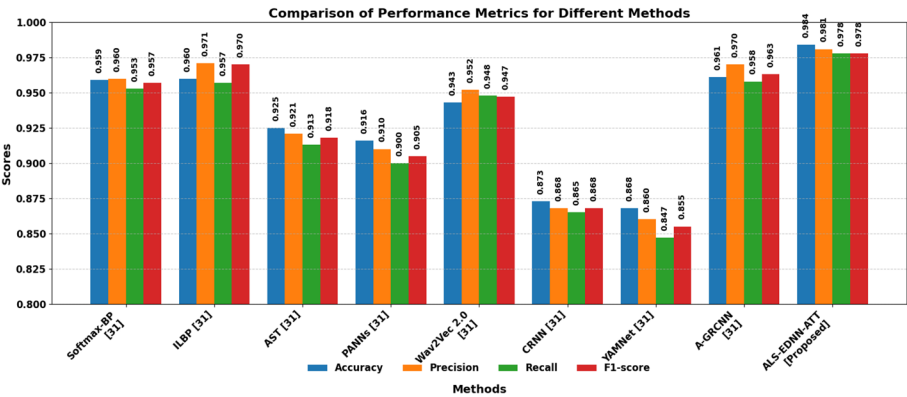


Fig. 22 Graphical comparison of existing and proposed methods

Table 9 Comparison of Inference, Training, and Resource Efficiency across Different Models for IPE Resource Recommendation

Metric	TTSM [32]	GCNs [32]	Transformer [32]	BERTs [32]	TF-IDF [32]	ALS-EDNN-ATT [proposed]
Inference time per query (ms)	28	35	40	45	10	8
Number of parameters (Million, M)	15.4	8.2	18.5	110	0.5	0.3
GPU memory usage (GB)	3.6	2.1	4.2	8.5	0.3	0.2
CPU inference time (ms)	180	220	260	350	60	55
Training time per epoch (s)	35	20	50	120	3	2
Total time for 5-fold cross-validation (h)	9	5	15	36	0.5	0.3

learning data, lacks deep feature representation capacity, and fails to dynamically adapt to diverse student behaviors in music education. ILBP [30] poorly generalizes high-level temporal dependencies, limiting performance prediction for sequential learning tasks. AST [30] requires large-scale data and high computational resources, and offers limited personalization for music learners. PANNs [30] frequently fail to adapt effectively to domain-specific nuances of student engagement, and practice patterns. Wav2Vec 2.0 [30] struggles to capture fine-grained music performance accuracy and is sensitive to noise and contextual variation. CRNN [30] suffers from vanishing gradients, high complexity, and lower precision in adapting dynamics. YAMNet [30] lacks robust contextual awareness, limiting the effectiveness in capturing detailed engagement and performance features essential for adaptive music education. A-GRCNN [30] suffers from slower convergence and limiting scalability for adaptive path recommendation. Thus, the proposed

Table 10 5-Fold Cross-Validation Performance of ALS-EDNN-ATT

Fold	Accuracy	Precision	Recall	F1-Score
1	0.981	0.978	0.975	0.976
2	0.986	0.984	0.981	0.982
3	0.983	0.981	0.979	0.980
4	0.985	0.982	0.980	0.981
5	0.984	0.981	0.978	0.979
Average	0.984	0.981	0.978	0.978
Std. Dev.	0.0018	0.0023	0.0021	0.0020

ALS-EDNN-ATT model overcomes these existing methods’ drawbacks by integrating deep enriched learning, attention-driven feature prioritization, and optimization, ensuring accurate, adaptive, and efficient personalized music learning path recommendations. From a theoretical perspective, this limitation restricts personalization to predictive accuracy rather than holistic learning trajectory optimization, preventing a unified integration of learner modeling and adaptive decision-making. Practically, the absence of dynamic sequencing and difficulty adjustment leads to inefficient practice, reduced learner engagement, and increased instructor intervention, particularly in formal music education settings where learners exhibit diverse abilities and progression rates. Addressing this gap is essential for developing scalable, adaptive, and domain-aware personalized music education systems capable of supporting sustained skill development and long-term learner engagement.

Traditional methods like TTSM [31], GCN [31], BERT [31], and TF-IDF [31] face limitations in handling complex, high-dimensional, and context-sensitive data. TTSM suffers from high computation and overfitting; GCN [31] struggles with shallow neighborhood capture and over-smoothing; BERT requires large pre-training and faces domain adaptation challenges; TF-IDF ignores semantic context and generates sparse vectors. The proposed ALS-EDNN-ATT (addresses these issues by using ALS to optimize network parameters, enriched features to improve representation, and attention mechanisms to focus on relevant temporal, structural, or semantic information. This combination enhances accuracy, generalization, and computational efficiency across various data types.

The existing research uses red music audio and lyrics, along with dynamic learner profiles, as its primary data. It processes this data by extracting emotional features from audio via short-time Fourier transform and attention mechanisms, analyzing lyrical semantics, modeling learner knowledge with Transformers, and generating personalized recommendations using heterogeneous information networks and hierarchical reinforcement learning. While the model improves recommendation accuracy, educational relevance, and emotional engagement, its limitations include reliance on red music only, cultural specificity to Chinese IPE, high computational complexity, and potential sensitivity to sudden changes in learner behavior. The proposed ALS-EDNN-ATT overcomes previous limitations by using ALS for optimized parameter tuning and attention mechanisms to capture both semantic and emotional features. It efficiently models learner profiles, achieving higher recommendation accuracy, relevance, and emotional engagement while remaining lightweight and scalable.

Although the ALS-EDNN-ATT model demonstrates strong performance, it is evaluated on a moderate-sized, structured dataset from formal music education settings, which may limit generalizability to informal or culturally diverse learning contexts. The

model focuses on learning path sequencing, difficulty, and pacing, and does not explicitly account for emotional expression, creativity, or socio-cultural influences that are central to music education. From an ethical perspective, reliance on learner data introduces risks of bias, privacy concerns, and reduced transparency in decision-making. In real-world deployment, challenges include computational overhead, integration with existing curricula, teacher acceptance, and long-term sustainability. Addressing these issues requires fairness-aware design, explainable AI mechanisms, strong data governance, and human-in-the-loop deployment strategies.

In this study, musical ability is conceptualized as a multidimensional construct, influenced by several factors, including prior musical knowledge, engagement, and skill mastery. The indicators selected for analysis such as pitch accuracy, rhythm stability, and performance accuracy were chosen based on their direct relevance to the goals of personalized music education, with a focus on measurable attributes that directly contribute to skill development. While factors like native language experience and musical enculturation are acknowledged as important influences, they were not incorporated due to the study's specific focus on formal music training and the limitations of the available dataset. The exclusion of these variables is noted as a limitation, which could affect the generalizability of the findings to broader populations, such as those from diverse linguistic and cultural backgrounds. Future research could address this gap by including a more comprehensive set of variables, enabling a more nuanced understanding of the interaction between cultural and linguistic influences and musical ability. This choice of indicators and the exclusion of certain variables, while enhancing the study's focus and feasibility, may limit the interpretation of results in a broader context. It is important to note that the results are most applicable to the specific educational setting and learner populations studied, and generalizing beyond these contexts should be approached with caution.

5 Conclusion

The ALS-EDNN-ATT model was developed to enable personalized music instruction through dynamic sequencing, adaptive difficulty, and tailored pacing in vocal and instrumental training. Using EDNN, it predicts learner behavior; ATT weighs features based on domain knowledge; and ALS optimization adapts learning paths according to engagement, performance, and feedback. Learners are classified as novice, intermediate, or advanced using domain-specific thresholds, framing personalization as a multiclass classification problem while maintaining distinct learning trajectories. Statistical validation with one-way and two-way ANOVA confirmed significant improvements in learning outcomes, including prior knowledge, engagement, skill metrics, module sequencing, task difficulty, and practice/feedback, demonstrating effective personalized learning. This comparative evaluation serves as a model-oriented validation by demonstrating that the proposed ALS-EDNN-ATT framework not only achieves statistically significant outcomes but also consistently outperforms state-of-the-art deep learning models in predictive accuracy, robustness, and adaptability for personalized music learning.

The research limitations include a purposeful focus on learning route sequencing, material difficulty, and tempo, while omitting emotive or socio-cultural factors. Additionally, it relies on a dataset of formal music training, which restricts applicability and long-term longitudinal research. Future endeavors will further personalization

by including emotional engagement, creativity-sensitive feedback, multimodal inputs, human-in-the-loop mentoring, and extensive multi-institutional datasets to augment scalability, educational significance, and longitudinal validation.

Abbreviations

DL	Deep Learning
ALS-EDNN-ATT	Artificial Lizard Search-driven Enriched deep neural networks with Attention mechanism
EDNN	Enriched deep neural networks
ALSO	Artificial Lizard Search Optimization
AI	Artificial Intelligence
DTSS	Definitive Teaching Support System
RNN	Recurrent Neural Network
RL	Reinforcement Learning
QES	Quantitative Evaluation System
AG	Automatic Generation
TDRL	Temporal-Difference Reinforcement Learning
IIMT	Intelligent Interactive Music Teaching
VR	Virtual Reality
CC	Cloud Computing
FCM	Fuzzy C-mean
EAs	Evolutionary Algorithms
RNN-LSTM	Recurrent Neural Network-Long Short-Term Memory
GA	Genetic Algorithm
CNN	Convolutional Neural Network
AIMS	Adaptive Instructive Suggestion Method
BPM	Beats Per Minute
MIDI	Musical Instrument Digital Interface
DNN	Deep Neural Networks
EDNN-ATT	Enriched deep neural networks with Attention mechanism
FFNN	Feedforward Neural Network
SNN	Shallow Neural Network
Att	Attention mechanism
Softmax-BP	Softmax Back Propagation
ILBP	Improved local binary pattern
AST	Audio Spectrogram Transformer
PANNS	Pretrained Audio Neural Networks
CRNN	Convolutional Recurrent Neural Network
YAMNet	Yet Another Music Recognition Network
A-GRCNN	Attention mechanism-gated recurrent convolutional neural network
LSTM	Long Short-Term Memory
Bi-LSTM	Bi56directional-Long Short-Term Memory
Att-GRU	Attention-Gated Recurrent Unit
CRM	Course Recommendation Model
AUC	Area Under the Curve
RMSE	Root Mean Square Error
FL-DTSS	Fuzzy Logic-driven Definitive Teaching Support System
Aoi	Areas of Interest
ELM	Extreme Learning Machine
ANOVA	Analysis of Variance
MANOVA	Multivariate Analysis of Variance
TISM	Transformer-based temporal sequence modeling
GCN	Graph Convolutional Networks
BERT	Bidirectional Encoder Representations from Transformers
TF-IDF	Term Frequency-Inverse Document Frequency

Author contributions

L. writing original draft preparation & methodology, L. investigation & writing review and editing.

Funding

No funding project.

Data availability

All data generated or analysed during this study are included in this article.

Declarations

Ethics approval and Consent to Participate

Not applicable. The study uses data that are fully anonymized and publicly accessible.

Consent for publication

Not applicable. The dataset used in this research does not include identifiable personal data, images, or recordings of individuals. All results are reported in aggregated form, ensuring participant anonymity.

Competing interests

The authors declare no competing interests.

Received: 4 November 2025 / Accepted: 4 March 2026

Published online: 31 March 2026

References

- Zheng J, Zhang Y, Zhang S. Audio-visual aesthetic teaching methods in college students' vocal music teaching by deep learning. *Sci Rep.* 2024;14(1):29386. <https://doi.org/10.1038/s41598-024-80640-7>.
- Wang K, Zhang P. Enhancing opera vocal education through advanced machine learning algorithms: analytics for talent development and curriculum design. *Interact Learn Environ.* 2025;33(1):1–21. <https://doi.org/10.1080/10494820.2025.2488991>.
- Jia Y. Impact of music teaching on student mental health using IoT, recurrent neural networks, and big data analytics. *Mob Networks Appl.* 2024;29(1):1–20. <https://doi.org/10.1007/s11036-024-02366-0>.
- Zhang M. Advancing deep learning for expressive music composition and performance modeling. *Sci Rep.* 2025;15(1):28007. <https://doi.org/10.1038/s41598-025-13064-6>.
- Aoonlamai N, Kwangmuang P. Integrating digital tools and constructivist learning: a ubiquitous learning framework for enhancing creativity in music education. *BMC Psychol.* 2025;13(1):1–18. <https://doi.org/10.1186/s40359-025-03300-z>.
- Cheng Y. Multimedia programmes: creating computer music to enhance creative thinking. *Interact Learn Environ.* 2023;31(10):6497–508. <https://doi.org/10.1080/10494820.2022.2039950>.
- Gao B, Li Q. Social entertainment robot based on neural network algorithm in personalized music course simulation. *Entertainment Comput.* 2025;52:100771. <https://doi.org/10.1016/j.entcom.2024.100771>.
- Yang X, Li Y. Long-term intervention through online courses in music education: Impact on assessment, performance, creativity, and musical culture. *Acta Psychol.* 2025;259:105363. <https://doi.org/10.1016/j.actpsy.2025.105363>.
- Song L. Design and implementation of remote piano teaching based on attention-Induced Multi-head Convolutional neural network optimized with Hunter–Prey optimization. *Int J Comput Intell Syst.* 2024;17(1). <https://doi.org/10.1007/s44196-023-00379-3>.
- You T, He H, Yue Y. Teacher Support and Pre-Service Preschool Teachers' Piano Skill: The Chain Mediation Effects of Music Self-Efficacy and Learning Engagement. *Behav Sci.* 2025;15(4):484. <https://doi.org/10.3390/bs15040484>.
- Wang X. Design of vocal music teaching system platform for music majors based on artificial intelligence. *Wirel Commun Mob Comput.* 2022;2022(1):5503834. <https://doi.org/10.1155/2022/5503834>.
- Jiang R, Mou X. 2024. The Analysis of Multi-track Music Generation with Deep Learning Models in Music Production Process. *IEEE Access*, 12, pp. 110322–110330. <https://doi.org/10.1109/ACCESS.2024.3439989>.
- Chen S. The application of big data and fuzzy decision support systems in the innovation of personalized music teaching in universities. *Int J Comput Intell Syst.* 2024;17(1):215. <https://doi.org/10.1007/s44196-024-00623-4>.
- Wang X. Deep learning-based personalized learning recommendation system design for T++ Guzheng Pedagogy. *Int J Inform Technol.* 2024;16(5):2775–81. <https://doi.org/10.1007/s41870-024-01871-5>.
- Chen Y, Sun Y. The usage of artificial intelligence technology in music education system under deep learning. *IEEE Access.* 2024;12:130546–56. <https://doi.org/10.1109/ACCESS.2024.3459791>.
- Peng Y. The analysis of optimization in music aesthetic education under artificial intelligence. *Sci Rep.* 2025;15(1):11545. <https://doi.org/10.1038/s41598-025-96436-2>.
- Su Y, Wang Y. Optimization of music education strategy guided by the temporal-difference reinforcement learning algorithm. *Soft Comput.* 2024;28(13):8279–91. <https://doi.org/10.1007/s00500-024-09631-0>.
- Han Y, Han L, Zeng C, Zhao W. The innovation path of VR technology integration into music classroom teaching in colleges and universities. *Sci Rep.* 2025;15(1):12200. <https://doi.org/10.1038/s41598-025-97003-5>.
- Fang Q. Research on VR e-learning based on load balancing algorithm in intelligent music education system. *Entertainment Comput.* 2024;50:100712. <https://doi.org/10.1016/j.entcom.2024.100712>.
- Jiang L. A fuzzy clustering approach for cloud-based personalized distance music education and resource management. *Soft Comput.* 2024;28(2):1707–24. <https://doi.org/10.1007/s00500-023-09525-7>.
- Dong L. Using deep learning and genetic algorithms for melody generation and optimization in music. *Soft Comput.* 2023;27(22):17419–33. <https://doi.org/10.1007/s00500-023-09135-3>.
- Li D, Liu Z. A big data dynamic approach for adaptive music instruction with deep neural fuzzy logic control. *EURASIP J Audio Speech Music Process.* 2025;2025(1):4. <https://doi.org/10.1186/s13636-025-00391-9>.
- Du H, Butkaew P. A deep learning-based intelligent curriculum system for enhancing public music education: a case study across three universities in Southwest China. *Sci Rep.* 2025;15(1):42798. <https://doi.org/10.1038/s41598-025-27153-z>.
- Han Y. Exploring a digital music teaching model integrated with recurrent neural networks under artificial intelligence. *Sci Rep.* 2025;15:7495. <https://doi.org/10.1038/s41598-025-92327-8>.
- Xia R, Li J, Li H. The construction of student-centered artificial intelligence online music learning platform based on deep learning. *Sci Rep.* 2025;15(1):15539. <https://doi.org/10.1038/s41598-025-95729-w>.
- Liu J, Liu J, Yan H. From Gaze to Music: AI-Powered Personalized Audiovisual Experiences for Children's Aesthetic Education. *Behav Sci.* 2025;15(12):1684. <https://doi.org/10.3390/bs15121684>.
- Cheng M, Zhang Y, Zhang W. Application and research of machine learning algorithms in personalized piano teaching system. *Int J High Speed Electron Syst.* 2025;34(02):2440094. <https://doi.org/10.1142/S0129156424400949>.
- Sham NSN, Ismail MJ, Sim YE. (2022). Perceptions toward music education. *Environment-Behaviour Proceedings Journal*, 7(S19), 353–357. <https://doi.org/10.21834/ebpj.v7iS19.3946>.
- Ismail MJ, Anuar AF, Loo FC. From physical to virtual: A new learning norm in music education for gifted students. *Int Rev Res Open Distrib Learn.* 2022;23(2):44–62.
- Chang D. Vocal performance evaluation of the intelligent note recognition method based on deep learning. *Sci Rep.* 2025;15(1):13927. <https://doi.org/10.1038/s41598-025-99357-2>.

31. Zhang L. Intelligent deep learning model for recommending ideological and political music education resources. *Sci Rep.* 2025;15(1):36402. <https://doi.org/10.1371/journal.pone.0297073>.
32. Haumann NT, Vuust P, Bertelsen F, Garza-Villarreal EA. Influence of musical enculturation on brain responses to metric deviants. *Frontiers in Neurosci.* 2018;12:218. <https://doi.org/10.3389/fnins.2018.00218>.
33. Zhang L, Xie S, Li Y, Shu H, Zhang Y. Perception of musical melody and rhythm as influenced by native language experience. *J Acoust Soc Am.* 2020;147(5):EL385–90. <https://doi.org/10.1121/10.0001179>.
34. Wang X, Ren X, Wang S, Yang D, Liu S, Li M, Yang M, Liu Y, Xu Q. Validation and applicability of the music ear test on a large Chinese sample. *PLoS ONE.* 2024;19(2):e0297073. <https://doi.org/10.1371/journal.pone.0297073>.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.